

SPECTRAL LIE ALGEBRAS AND BAR TOWERS

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ABSTRACT. We compare two constructions of higher enveloping algebras in a presentably symmetric monoidal stable ∞ -category. Chevalley–Eilenberg chains preserve finite products, and the resulting Lie-theoretic cone along actual bar is equivalent to the cone obtained by Koszul dualizing the inclusions of little-disks operads. For $T(h)$ -local spectra, with $T(0) = H\mathbb{Q}$, we then prove that the induced functor from spectral Lie algebras to compatible bar towers is an equivalence exactly when $h = 0$ or $h = 1$. At every height $h \geq 2$ and every prime, connective covers of $K(1)$ give a nonzero bar tower annihilated by the right adjoint. The rational case follows from the comparison and also admits an independent proof by free suspensions and shuffle bar.

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1. INTRODUCTION

Higher enveloping algebras associate to a spectral Lie algebra a compatible sequence of nonunital little-disks algebras. Two constructions of this sequence are available. One applies Chevalley–Eilenberg chains to internal loop objects and uses ordinary bar for the transition functors. The other applies operadic Koszul duality to the inclusions of little-disks operads. Shi asks whether the resulting diagrams agree [8, Conjecture 2.4.10], and whether every compatible bar tower arises from a spectral Lie algebra at each positive chromatic height [8, Conjecture 2.4.15]. We prove the comparison and determine exactly when the inverse-limit functor is an equivalence.

Let $\mathcal{C}$ be a presentably symmetric monoidal stable ∞ -category; in particular, its tensor product preserves colimits separately in each variable. Write Lie for the spectral Lie operad, namely the

operadic Koszul dual of the reduced commutative operad, and let

$$\mathrm{CE}: \mathrm{Alg}_{\mathrm{Lie}}(\mathcal{C}) \longrightarrow \mathrm{Coalg}_{\mathbb{E}_\infty}^{\mathrm{coaug}}(\mathcal{C})$$

denote Chevalley–Eilenberg chains with their ordinary counital, coaugmented cocommutative coalgebra structure. For $n \geq 0$, write

$$B_{n+1}: \mathrm{Alg}_{\mathbb{E}_{n+1}}^{\mathrm{nu}}(\mathcal{C}) \longrightarrow \mathrm{Alg}_{\mathbb{E}_n}^{\mathrm{nu}}(\mathcal{C})$$

for one-fold reduced bar, with its residual $\mathbb{E}_n$ -algebra structure. Here $\mathrm{Alg}_{\mathbb{E}_0}^{\mathrm{nu}}(\mathcal{C}) \simeq \mathcal{C}$. The Lie-theoretic construction gives functors U_n and coherent equivalences $B_{n+1}U_{n+1} \simeq U_n$ whenever the finite-product comparison for CE is an equivalence. The following theorem supplies that input and identifies the resulting cone.

Theorem 1.1. *For every presentably symmetric monoidal stable ∞ -category $\mathcal{C}$:*

- (i) *The functor CE preserves finite products. Consequently its underlying functor, given by the tensor unit plus Lie indecomposables, is symmetric monoidal from the cartesian structure on Lie algebras to the tensor structure on augmented objects of $\mathcal{C}$.*
- (ii) *The Lie-theoretic higher enveloping functors define a cone*

$$\mathrm{Alg}_{\mathrm{Lie}}(\mathcal{C}) \longrightarrow (\cdots \xrightarrow{B_3} \mathrm{Alg}_{\mathbb{E}_2}^{\mathrm{nu}}(\mathcal{C}) \xrightarrow{B_2} \mathrm{Alg}_{\mathbb{E}_1}^{\mathrm{nu}}(\mathcal{C}) \xrightarrow{B_1} \mathcal{C})$$

of presentable categories and colimit-preserving functors.

- (iii) *This entire cone is equivalent to the cone obtained by Koszul dualizing the inclusions $\mathbb{E}_n \rightarrow \mathbb{E}_{n+1}$ and transporting along little-disks self-duality. The comparison includes the horizontal arrows and their compatibility with the Lie vertex.*

Part (i) removes the additional assumption in [8, §2.4.6]; part (iii) proves Conjecture 2.4.10 of that paper. Changing the little-disks self-duality conjugates the comparison by operadic automorphisms at the vertices, as described in [theorem 7.2](#).

We next specialize to chromatic homotopy theory. Fix a prime p . For $h \geq 1$, let $T(h)$ be the telescope of a periodic self-map of a finite p -local spectrum of type h , and put $T(0) = H\mathbb{Q}$. Define

$$\mathrm{BarTower}_h = \varprojlim_{n \geq 0} (\mathrm{Alg}_{\mathbb{E}_n}^{\mathrm{nu}}(\mathrm{Sp}_{T(h)}), B_{n+1}).$$

The limit is taken in ∞ -categories, equivalently in Pr^{L} for this diagram. Its degree-zero projection is denoted by $\mathrm{pr}_0: \mathrm{BarTower}_h \rightarrow \mathrm{Sp}_{T(h)}$. The cone of [theorem 1.1](#) induces an adjunction

$$(1.1) \quad U_\infty: \mathrm{Alg}_{\mathrm{Lie}}(\mathrm{Sp}_{T(h)}) \rightleftarrows \mathrm{BarTower}_h: T_\infty.$$

At positive heights, U_∞ is fully faithful by [8, Theorem 2.3.1]. The next theorem determines its essential surjectivity.

Theorem 1.2. *For every prime p the following statements hold.*

- (i) *At heights $h = 0$ and $h = 1$, the functor U_∞ in (1.1) is an equivalence.*
- (ii) *At every height $h \geq 2$, there is $A \in \mathrm{BarTower}_h$ with*

$$A_0 \neq 0, \quad \mathrm{oblv}_{\mathrm{Lie}} T_\infty(A) \simeq 0.$$

In particular, U_∞ is not essentially surjective.

Thus Conjecture 2.4.15 of [8] holds exactly at height one among the positive heights, and its rational analogue also holds.

Let I denote the augmentation-ideal functor and $K(1)$ height-one Morava K -theory at p . For every $h \geq 2$ and a sufficiently large integer c , the counterexample has terms

$$(1.2) \quad A_n \simeq \mathrm{res}_{\mathbb{E}_n}^{\mathbb{E}_\infty} I\left(L_{T(h)} \Sigma_+^\infty \Omega^\infty \tau_{\geq c} \Sigma^{-n} K(1)\right), \quad n \geq 0.$$

The multiplication is induced by the additive infinite loop space of the spectrum.

The comparison and its consequences. The two theorems have distinct proof mechanisms. To prove the comparison, we first filter Lie algebras so that the associated graded Chevalley–Eilenberg complex is a symmetric coalgebra. On each mixed symmetric power, the canonical product comparison is a subgroup transfer followed by a specified coset projection. This composite is the identity by the induction–coinduction triangle identity. Conservativity of associated graded on nonnegative filtrations then proves finite-product preservation.

The remaining comparison is proved in positive-arity symmetric sequences. In this category only finitely many tensor lengths contribute in each arity, and the relevant permutation actions are induced. These two properties identify ordinary coalgebras with coalgebras for the operadic comonad and make enhanced bar an equivalence. Recovering its comonad gives a normalized, coherent little-disks self-duality. Under that normalization, the enhanced bars of both enveloping constructions are the same restriction of CE. Right substitution retains the operadic bimodule actions, which carry this comparison to an arbitrary $\mathcal{C}$ through geometric realizations.

At height zero, the comparison transports Shi’s rational equivalence for the Koszul-dual cone [8, Theorem 2.4.12] to actual bar towers. Appendix A also gives a direct proof, independent of the comparison theorem. It proves that the suspension of any rational bar tower is free and computes the adjunction unit by a shuffle-bar calculation.

At positive heights, we use the loop functor and costabilization of the actual bar tower. Periodicity forces mapping spaces into the kernel of $\mathrm{oblv}_{\mathrm{Lie}} T_\infty$ to be null with respect to a finite type- h complex. These mapping spaces are also local for mod- p homology equivalences. At height one a Moore space supplies a uniform truncation bound, and conservativity of loops makes the kernel vanish. At heights $h \geq 2$, the localized group-ring construction on connective covers preserves finite colimits on $\mathrm{thick}(K(1))$. It therefore produces the compatible tower (1.2). Finite type- h spectra force its image under the right adjoint to vanish, whereas Bousfield’s example detects the nonzero zeroth term.

Conventions. All categories, operads, adjunctions, and diagrams are ∞ -categorical. Tensor products, composition products, realizations, orbits, and fixed points are derived. A reduced spectral operad P has $P(0) = 0$ and $P(1) = S$. We use the spectral normalization of Lie and the actual-bar construction of U_n from [8, §2.2]. The notation Map denotes a mapping space and map a mapping spectrum. Suspension in a pointed category $\mathcal{X}$ is denoted by $\Sigma_{\mathcal{X}}$; otherwise suspension is spectral or that of pointed spaces, as determined by its argument. In $\mathrm{Sp}_{T(h)}$ the unit is $L_{T(h)} S_{(p)}$ and tensor product is localized smash product. For a pointed space Y , the notation $\Sigma^\infty Y$ means its reduced suspension spectrum.

2. OPERADIC BAR AND COALGEBRA NORMALIZATION

Write $\mathbb{I}$ for the identity operad, concentrated in arity one. The operads used here are

$$P_n = \Sigma_+^\infty \mathbb{E}_n^{\mathrm{uu}}, \quad P_0 = \mathbb{I}, \quad \mathrm{Com}(r) = S \quad (r \geq 1).$$

The standard inclusions and their compatible maps to Com are denoted by

$$\iota_n : P_n \longrightarrow P_{n+1}, \quad i_n : P_n \longrightarrow \mathrm{Com}.$$

The operadic bar construction and Koszul dual are

$$BP = B(\mathbb{I}, P, \mathbb{I}), \quad KP = (BP)^\vee, \quad (-)^\vee = \mathrm{map}(-, S),$$

where duality is taken aritywise. We use the spectral normalization $\mathrm{Lie} = K\mathrm{Com}$; thus Lie is the shifted spectral Lie operad.

For $a \in \mathbb{Z}$, our operadic suspension convention is

$$(2.1) \quad (\mathfrak{s}^a P)(r) = S^{a\bar{p}_r} \otimes P(r), \quad (\mathfrak{s}^a P) \circ X \simeq \Sigma^{-a}(P \circ \Sigma^a X),$$

where $\bar{\rho}_r$ is the reduced permutation representation. The corresponding equivalence of algebra categories is

$$S_a : \text{Alg}_P(\mathcal{C}) \xrightarrow{\simeq} \text{Alg}_{s^a P}(\mathcal{C}), \quad |S_a A| = \Sigma^{-a} |A|.$$

We use the same notation for cooperad coalgebras. Conjugation by the invertible unary symmetric sequence $\Sigma^a \mathbb{I}$ supplies coherent identifications, including the symmetric-group actions,

$$(2.2) \quad B(s^a P) \simeq s^a B P, \quad K(s^a P) \simeq s^{-a} K P.$$

For a reduced operad P , write

$$\text{indec}_P = \mathbb{I} \circ_P - : \text{Alg}_P(\mathcal{C}) \rightleftarrows \mathcal{C} : \text{triv}_P.$$

Its enhanced operadic bar is denoted by $\mathbf{B}_P$. This enhancement takes values in ind-conilpotent divided-power BP -coalgebras and is the comparison functor associated to the comonad $\text{indec}_P \text{triv}_P$. Ordinary coalgebras will mean coalgebras without divided-power structure. For a unital little-disks operad, $\text{Coalg}_{\mathbb{E}_n}^{\text{coaug}}$ denotes ordinary counital, coaugmented coalgebras. An augmented object is a retraction $\mathbf{1} \rightarrow X \rightarrow \mathbf{1}$. Nonunital $\mathbb{E}_n$ -algebras correspond to augmented $\mathbb{E}_n$ -algebras by unitization and augmentation ideal.

Lemma 2.1 (Enhanced bar functoriality). *Let P be a reduced spectral operad.*

(i) *There are natural identifications*

$$\text{indec}_P(A) \simeq |B_\bullet(\mathbb{I}, P, A)|, \quad \text{indec}_P \text{triv}_P \simeq B P \circ -.$$

The latter is an identification of comonads, and the induced coaction on $\text{indec}_P(A)$ is the enhancement $\mathbf{B}_P(A)$.

(ii) *For a map $f : P \rightarrow Q$ of reduced operads, there is a natural equivalence of enhanced coalgebras*

$$(2.3) \quad \mathbf{B}_Q(f_! A) \simeq (Bf)_* \mathbf{B}_P(A), \quad f_! A = Q \circ_P A.$$

Here $(-)_$ denotes corestriction of coalgebra structure. The associated map of comonads is the mate*

$$(2.4) \quad \text{indec}_P \text{triv}_P \simeq \text{indec}_Q f_! f^* \text{triv}_Q \xrightarrow{\text{indec}_Q \varepsilon \text{triv}_Q} \text{indec}_Q \text{triv}_Q.$$

(iii) *Enhanced bar and extension of scalars commute with colimit-preserving symmetric monoidal functors $G : \mathcal{C} \rightarrow \mathcal{C}'$. So does passage from divided-power coalgebras to ordinary coalgebras by the norm construction, including adjoining a counit and coaugmentation. These comparisons are compatible with maps of operads and with composition of the functors G .*

Proof. The comonad identification in (i) is [4, Propositions 6.2 and 6.5]. Its comultiplication is $\text{indec}_P \eta \text{triv}_P$, where η is the unit of the indecomposables–trivial-algebra adjunction. Splitting the two-sided bar at an intermediate level identifies this map with the cooperad decomposition of BP .

For (ii), associativity of relative composition gives

$$\mathbb{I} \circ_Q (Q \circ_P A) \simeq \mathbb{I} \circ_P A.$$

The identifications $\text{indec}_P \simeq \text{indec}_Q f_!$ and $\text{triv}_P \simeq f^* \text{triv}_Q$ give the mate (2.4). Splitting a bar at an intermediate level identifies this mate with Bf . The same splitting with last input A proves (2.3). Splitting at several levels gives its compatibility with iterated coactions. Equivalently, these compatibilities are the coherent unit and counit identities for the adjunctions in (2.4).

Each term of the two-sided bar is formed from tensor products, coproducts, and homotopy orbits. These constructions and geometric realization commute with G . The bar splittings therefore also commute with G , which proves the assertion for the enhanced structure. For the underlying functoriality of divided-power coalgebra categories under colimit-preserving oplax monoidal functors, see [4, Appendix B.2].

To include ordinary coalgebras, let H be a finite group and let $Z \in \text{Fun}(BH, \mathcal{C})$. The norm maps fit into the commutative square

$$(2.5) \quad \begin{array}{ccc} G(Z_{hH}) & \xrightarrow{G(\text{Nm}_H)} & G(Z^{hH}) \\ \simeq \downarrow & & \downarrow \\ (GZ)_{hH} & \xrightarrow{\text{Nm}_H} & (GZ)^{hH}. \end{array}$$

The right vertical map is the comparison to a limit. The finite-free resolution defining the transfer norm constructs its finite-stage maps from finite sums, or equivalently from the identification of finite induction and coinduction, and finite fiber and cofiber constructions. An exact functor preserves these maps. Passing to the realization on the orbit side and the compatible cone on the fixed-point side gives (2.5). Applying this construction to composites of finite covers gives compatibility with composites of norms. Since G preserves colimits between stable categories, it is exact, so this argument applies. Thus norm-forgetting commutes with G . The divided-power and norm constructions used here are described in [4, §§2, 7 and Appendix A]. $\square$

Lemma 2.2 (Double duality). *Reduced operadic bar is an equivalence*

$$B : \text{Op}_{\text{red}}(\text{Sp}) \xrightarrow{\simeq} \text{Coop}_{\text{red}}(\text{Sp}).$$

If P is aritywise finite, then BP is aritywise finite, and there are natural equivalences of operads and cooperads, respectively,

$$\delta_P : P \xrightarrow{\simeq} K^2P, \quad d_P : BKP \xrightarrow{\simeq} P^\vee.$$

For a map $f : P \rightarrow Q$ between aritywise finite reduced operads,

$$(2.6) \quad d_P BK(f) \simeq f^\vee d_Q.$$

Proof. The reduced bar–cobar equivalence is [4, Theorem 3.4]. In a normalized bar tree with r labeled leaves, every nonidentity vertex has at least two inputs, so there are at most $r - 1$ such vertices. There are finitely many trees and nondegenerate levelings of this kind. Their cells are finite tensor products of terms of P , and an automorphism fixing every labeled leaf is the identity. Consequently the normalized bar in each arity is a finite cell construction from finite spectra, and hence is finite.

Termwise duality exchanges this finite bar with the corresponding cobar. It identifies K^2P with the operadic cobar of BP ; let δ_P be the resulting bar–cobar unit. Dualizing δ_P defines d_P , using $K^2P = (BKP)^\vee$. Naturality of the unit gives (2.6), with its coherences. In particular, since $\text{Com}^\vee = \text{Com}$, this fixes the equivalence

$$(2.7) \quad d_{\text{Com}} : \text{BLie} \xrightarrow{\simeq} \text{Com}.$$

$\square$

3. CHEVALLEY–EILENBERG CHAINS AND FINITE PRODUCTS

Using the normalization (2.7), define

$$(3.1) \quad \begin{aligned} \text{CE} : \text{Alg}_{\text{Lie}}(\mathcal{C}) &\xrightarrow{\mathbf{B}_{\text{Lie}}} \text{Coalg}_{\text{BLie}}^{\text{nil,dp}}(\mathcal{C}) \xrightarrow{(d_{\text{Com}})_*} \text{Coalg}_{\text{Com}}^{\text{nil,dp}}(\mathcal{C}) \\ &\longrightarrow \text{Coalg}_{\mathbb{E}_\infty}^{\text{coaug}}(\mathcal{C}). \end{aligned}$$

Here nil, dp denotes ind-conilpotent divided-power coalgebras. The last functor uses finite-group norms and adjoins a counit and coaugmentation. Its underlying augmented object is

$$(3.2) \quad F(\mathfrak{g}) = \mathbf{1} \oplus \text{indec}_{\text{Lie}}(\mathfrak{g}).$$

This is the Chevalley–Eilenberg construction used in [8, §2.4.6].

Theorem 3.1. *For every presentably symmetric monoidal stable ∞ -category $\mathcal{C}$, the projection-induced maps*

$$\mathrm{CE}(\mathfrak{g} \times \mathfrak{h}) \longrightarrow \mathrm{CE}(\mathfrak{g}) \otimes \mathrm{CE}(\mathfrak{h}), \quad \mathrm{CE}(0) \longrightarrow \mathbf{1}$$

are equivalences of ordinary coaugmented cocommutative coalgebras. Consequently $F = \mathbf{1} \oplus \mathrm{indec}_{\mathrm{Lie}}$ is symmetric monoidal from the cartesian monoidal structure on $\mathrm{Alg}_{\mathrm{Lie}}(\mathcal{C})$ to the tensor product of augmented objects.

We prove the theorem by filtering the canonical comparison map. Its associated graded is the exponential comparison for cofree ind-conilpotent divided-power coalgebras after applying norms. The next lemma identifies that comparison as a map, and the nonnegative filtration then detects its equivalence.

Lemma 3.2 (The exponential coalgebra comparison). *Let*

$$\mathrm{Sym}^c(X) = \bigoplus_{r \geq 0} (X^{\otimes r})_{h\Sigma_r}$$

carry the ordinary coaugmented cocommutative coalgebra structure obtained from the cofree ind-conilpotent divided-power Com-coalgebra by norms. The comparison induced by the two projections is an equivalence

$$(3.3) \quad \mathrm{Sym}^c(X \oplus Y) \longrightarrow \mathrm{Sym}^c(X) \otimes \mathrm{Sym}^c(Y).$$

Proof. Distributing tensor products over direct sums gives, in total weight r ,

$$(3.4) \quad ((X \oplus Y)^{\otimes r})_{h\Sigma_r} \simeq \bigoplus_{a+b=r} (X^{\otimes a})_{h\Sigma_a} \otimes (Y^{\otimes b})_{h\Sigma_b}.$$

For $a+b=r$, set $H = \Sigma_a \times \Sigma_b \subseteq \Sigma_r = G$ and $M = X^{\otimes a} \otimes Y^{\otimes b}$. Before taking orbits, the summand with a occurrences of X and b occurrences of Y is $\mathrm{Ind}_H^G M$. Its (a, b) coproduct component, followed by the two projections, is

$$(3.5) \quad M_{hH} \simeq (\mathrm{Ind}_H^G M)_{hG} \xrightarrow{\mathrm{tr}_H^G} (\mathrm{Res}_H^G \mathrm{Ind}_H^G M)_{hH} \xrightarrow{\mathrm{pr}} M_{hH}.$$

This composite is the identity by an adjunction triangle. Indeed, put $R = \mathrm{Res}_H^G$ and $J = \mathrm{Ind}_H^G \simeq \mathrm{Coind}_H^G$. For the adjunction $R \dashv J$, with unit η and counit ε , one has

$$(3.6) \quad (JM \xrightarrow{\eta_{JM}} JRJM \xrightarrow{J\varepsilon_M} JM) \simeq \mathrm{id}_{JM}.$$

Taking G -orbits and applying Shapiro's equivalence identifies this composite with (3.5). The counit $\varepsilon_M : RJM \rightarrow M$ is projection to the distinguished coset H . Every other coset places a Y in an X -position or an X in a Y -position and is killed by the projections. Thus the triangle identity identifies the projection-induced comparison for arbitrary H -actions on M .

The coalgebra coproduct has precisely this transfer description: cooperadic decomposition records the unordered splitting and the norm records its order. If $a=0$ or $b=0$, the required assertion is the counit identity. Thus (3.3) is (3.4) in each weight. Tensor product preserves the direct sum over weights, proving the result. $\square$

Let $\mathbb{N} = \{0, 1, 2, \dots\}$ and equip

$$\mathrm{Fil}(\mathcal{C}) = \mathrm{Fun}(\mathbb{N}, \mathcal{C}), \quad \mathrm{Gr}(\mathcal{C}) = \mathrm{Fun}(\mathbb{N}^{\mathrm{disc}}, \mathcal{C})$$

with additive Day convolution. For a filtered object Z , put

$$(\mathrm{gr} Z)_0 = Z_0, \quad (\mathrm{gr} Z)_r = \mathrm{cofib}(Z_{r-1} \rightarrow Z_r) \quad (r > 0).$$

Lemma 3.3. *The functors gr and $\mathrm{colim} : \mathrm{Fil}(\mathcal{C}) \rightarrow \mathcal{C}$ preserve colimits and are symmetric monoidal. Moreover, gr detects equivalences.*

Proof. The cofiber formula proves colimit preservation for gr . The filtered tensor product is

$$(Z \otimes W)_r = \text{colim}_{i+j \leq r} Z_i \otimes W_j.$$

Its quotient by the preceding stage is, by the pushout-product formula,

$$\bigoplus_{i+j=r} (\text{gr}_i Z) \otimes (\text{gr}_j W).$$

These equivalences come from quotienting the boundary of the same multi-index diagram, so they respect permutations and all iterated products. One can also check their coherence on step objects $0 \rightarrow \cdots \rightarrow 0 \rightarrow X \xrightarrow{\text{id}} X \rightarrow \cdots$, which generate the filtered category under colimits. The Day formula and cofinality of addition $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ prove the symmetric monoidality of colimit.

If a filtered map is an equivalence on associated graded, it is an equivalence in degree zero. The cofiber sequence for consecutive filtration degrees proves the same assertion in every degree by induction. Thus associated graded is conservative. Compare [2, §5, especially Propositions 5.7 and 5.9]. $\square$

For $\mathfrak{g} \in \text{Alg}_{\text{Lie}}(\mathcal{C})$, let

$$(3.7) \quad \mathfrak{g}^f = (0 \longrightarrow \mathfrak{g} \xrightarrow{\text{id}} \mathfrak{g} \xrightarrow{\text{id}} \cdots)$$

be its filtered Lie algebra. On positive filtration indices, the r -ary operation is that of $\mathfrak{g}$, placed in the sum of the indices; an operation with a zero input is zero. Substitution adds these indices, so this defines the filtered algebra structure.

Lemma 3.4. *There are natural equivalences of Lie algebras*

$$\text{colim } \mathfrak{g}^f \simeq \mathfrak{g}, \quad \text{gr}(\mathfrak{g}^f) \simeq \text{triv}_{\text{Lie}}(|\mathfrak{g}| \langle 1 \rangle),$$

where $\langle 1 \rangle$ means placement in weight one, not suspension. Moreover, $(\mathfrak{g} \times \mathfrak{h})^f \simeq \mathfrak{g}^f \times \mathfrak{h}^f$.

Proof. The underlying associated graded is concentrated in weight one. An operation on $r \geq 2$ such inputs has output weight r , where the object is zero. The spaces of these operations and their coherences are therefore contractible; the unary operation is prescribed by the operadic unit. This identifies the associated graded as a trivial Lie algebra. The colimit assertion follows from the constant positive stages, and the product assertion follows because limits of operadic algebras are created on underlying objects. $\square$

Proposition 3.5 (The CE filtration). *There is a natural nonnegative filtered ordinary coaugmented cocommutative coalgebra $\mathcal{F}\text{CE}(\mathfrak{g})$ with*

$$(3.8) \quad \text{colim } \mathcal{F}\text{CE}(\mathfrak{g}) \simeq \text{CE}(\mathfrak{g})$$

and

$$(3.9) \quad \text{gr } \mathcal{F}\text{CE}(\mathfrak{g}) \simeq \text{Sym}^c(|\mathfrak{g}| \langle 1 \rangle).$$

Its weight-zero associated graded is $\mathbf{1}$, and its weight- r associated graded for $r > 0$ is $(|\mathfrak{g}|^{\otimes r})_{h\Sigma_r}$.

Proof. Apply (3.1) in $\text{Fil}(\mathcal{C})$ to (3.7). Lemmas 2.1, 3.3, and 3.4 give (3.8) and reduce (3.9) to computing $\text{CE}(\text{triv}_{\text{Lie}} V)$. The comonad identification and the fixed normalization d_{Com} give

$$\text{indec}_{\text{Lie}} \text{triv}_{\text{Lie}}(V) \simeq B\text{Lie} \circ V \simeq \bigoplus_{r \geq 1} (V^{\otimes r})_{h\Sigma_r}.$$

Under this identification of comonads, the coaction is that of the cofree ind-conilpotent divided-power coalgebra. Adjoining the counit and coaugmentation and applying norms gives $\text{Sym}^c(V)$ and proves (3.9). $\square$

Proof of Theorem 3.1. The tensor product of ordinary counital cocommutative coalgebras is their categorical product, with projections induced by the counits. This remains true after specifying coaugmentations. It is the dual of the coproduct formula for commutative algebras; see [7, Example 3.2.4.4 and Proposition 3.2.4.7]. The two Lie algebra projections therefore induce a canonical filtered coalgebra map

$$\mathcal{FCE}(\mathfrak{g} \times \mathfrak{h}) \longrightarrow \mathcal{FCE}(\mathfrak{g}) \otimes \mathcal{FCE}(\mathfrak{h}).$$

Since $|\mathfrak{g} \times \mathfrak{h}| \simeq |\mathfrak{g}| \oplus |\mathfrak{h}|$, Proposition 3.5 identifies its associated graded with the canonical comparison of Lemma 3.2, which is an equivalence. Lemma 3.3 now makes the filtered comparison an equivalence. Taking colimits and using (3.8) proves the binary-product assertion. Also $\text{indec}_{\text{Lie}}(0) = 0$, so $\text{CE}(0) = \mathbf{1}$.

A finite-product-preserving functor between cartesian monoidal categories has its canonical symmetric monoidal structure. The forgetful functor from ordinary coaugmented cocommutative coalgebras to augmented objects is symmetric monoidal from this cartesian structure to tensor product. Their composite is F , which proves the remaining assertion. $\square$

4. RIGHT SUBSTITUTION AND ORDINARY BAR MAPS

Let

$$\mathbf{SSeq} = \text{Fun}(\text{Fin}^\simeq, \text{Sp}), \quad \mathbf{SSeq}_{>0} = \{X \in \mathbf{SSeq} : X(\emptyset) = 0\}.$$

Here finite sets may be empty. The Day tensor product on $\mathbf{SSeq}$ is denoted by $\otimes_D$; its unit $\mathbf{1}_D$ is concentrated in arity zero. The composition product $\circ$ on positive symmetric sequences has unit $\mathbb{I}$, concentrated in arity one. For a nonempty finite set J and positive symmetric sequences $X_1, \dots, X_r$,

$$(4.1) \quad (X_1 \otimes_D \cdots \otimes_D X_r)(J) = 0 \quad (r > |J|),$$

because the Day formula is indexed by decompositions of J into r nonempty subsets.

Right substitution by $X \in \mathbf{SSeq}_{>0}$ is the functor $R_X(Y) = Y \circ X$. A functor between categories carrying these actions is called *right-linear* if it is a module functor for $(\mathbf{SSeq}_{>0}, \circ)$: its equivalences $T(Y \circ X) \simeq T(Y) \circ X$ include the unit, associativity, and naturality data. We use augmented algebras and coaugmented coalgebras in $\mathbf{SSeq}$ whose augmentation or coaugmentation ideals are positive. In particular,

$$\text{Alg}_{\mathbb{E}_n}^{\text{aug}}(\mathbf{SSeq})_{>0} \simeq \text{Alg}_{P_n}(\mathbf{SSeq}_{>0}).$$

Write B_{n+1} for one-direction reduced bar, viewed as a functor from P_{n+1} -algebras to P_n -algebras.

Proposition 4.1 (Ordinary bar maps). *There are maps of spectral operads*

$$\beta_{n+1} : P_{n+1} \longrightarrow \mathbf{s}P_n$$

such that, on positive symmetric sequences,

$$(4.2) \quad B_{n+1} \simeq S_1^{-1}(\beta_{n+1})_!$$

These are the bar maps of [5, Theorem 3.11]. The comparison is right-linear and satisfies

$$(4.3) \quad \text{indec}_{P_n} B_{n+1} \simeq \Sigma \text{indec}_{P_{n+1}}.$$

At $n = 0$, this gives $B_1 \simeq \Sigma \text{indec}_{P_1}$.

We recover the operad map from the action of bar on free algebras, using right substitution and the resulting bimodule kernel.

Lemma 4.2 (Right substitution). *For $X \in \mathbf{SSeq}_{>0}$, the functor R_X extends to a colimit-preserving strong symmetric monoidal functor on $(\mathbf{SSeq}, \otimes_D)$ and preserves positive symmetric sequences. Evaluation at $\mathbb{I}$ gives a monoidal equivalence*

$$(4.4) \quad \text{End}_{(\mathbf{SSeq}_{>0}, \circ)}(\mathbf{SSeq}_{>0}) \simeq (\mathbf{SSeq}_{>0}, \circ),$$

where the left side consists of right-module endofunctors.

Proof. The identification

$$(Y \otimes_D Z) \circ X \simeq (Y \circ X) \otimes_D (Z \circ X)$$

reindexes a partition of a finite set into blocks, each partitioned again. It respects further partitions and bijections, which supplies the symmetric monoidal coherence. Substitution is colimit-preserving in its left argument, and its arity-zero formula gives $R_X(\mathbf{1}_D) = \mathbf{1}_D$. Positivity is immediate from the same formula.

If T is a right-module endofunctor, its module constraint gives

$$T(X) \simeq T(\mathbb{I} \circ X) \simeq T(\mathbb{I}) \circ X.$$

Evaluation at $\mathbb{I}$ likewise identifies module natural transformations and their higher homotopies. Composition corresponds to the composition product, in the displayed order. In particular, right-linear comonads and their maps correspond to cooperads and cooperad maps. $\square$

Lemma 4.3 (Kernels of right-linear functors). *Let P, Q be reduced spectral operads, and suppose*

$$T : \text{Alg}_P(\text{SSeq}_{>0}) \longrightarrow \text{Alg}_Q(\text{SSeq}_{>0})$$

is right-linear and preserves geometric realizations. The object $N_T = T(P)$, where $P = \text{Free}_P(\mathbb{I})$ has its regular left P -structure, is naturally a (Q, P) -bimodule, and

$$(4.5) \quad T(A) \simeq N_T \circ_P A.$$

This construction respects right-linear natural transformations and coherent diagrams. If in addition $T \text{Free}_P \simeq \text{Free}_Q$ as right-linear functors, then $T \simeq f_!$ for an operad map $f : P \rightarrow Q$ determined by this data.

Proof. The left Q -action is part of $T(P)$. Its right P -action is

$$N_T \circ P \simeq T(P \circ P) \xrightarrow{T(\mu_P)} T(P).$$

The unit and associativity of P give the right-action identities. This is a map of left Q -algebras, so the two actions commute. Right linearity also gives

$$T(\text{Free}_P X) = T(P \circ X) \simeq N_T \circ X.$$

Apply T to the simplicial free-algebra resolution of A . Its image is the two-sided bar construction $B_\bullet(N_T, P, A)$: the face that substitutes in P is precisely the right action just defined. Realization proves (4.5), including its left Q -structure. The construction is functorial for transformations and diagrams.

Under the additional hypothesis, $N_T \simeq Q$ as a left Q -algebra. By the free-left-algebra adjunction a compatible action $Q \circ P \rightarrow Q$ is determined by a map $P \rightarrow Q$. Its right-action identities are the operad identities for this map. Thus $T \simeq Q \circ_P - = f_!$. $\square$

Proof of Proposition 4.1. The augmented bar construction of [7, §§5.2.2–5.2.3] has a natural free-algebra comparison

$$(4.6) \quad B_{n+1} \text{Free}_{P_{n+1}} \simeq \text{Free}_{P_n} \Sigma.$$

It follows by viewing an $\mathbb{E}_{n+1}$ -algebra as an associative algebra internal to $\mathbb{E}_n$ -algebras and applying the free-algebra suspension formula for bar. This is the free triangle in the proof of [5, Theorem 3.11, using Theorem 2.4]. The construction applies in SSeq and preserves positivity.

Bar preserves geometric realizations. Its construction and (4.6) commute with right substitution by Lemma 4.2 and enhanced-bar naturality [7, Example 5.2.3.11]. Apply Lemma 4.3 to $S_1 B_{n+1}$ to obtain β_{n+1} and (4.2). For a spectrum Z , the functor

$$Y \longmapsto \bigoplus_{r \geq 0} (Y(r) \otimes Z^{\otimes r})_{h\Sigma_r}$$

is colimit-preserving and Day symmetric monoidal. It carries (4.6) to the same free triangle in spectra. The full-faithfulness statement for little-disks operads and their suspensions in [5, Theorem 3.8] identifies the recovered map with the bar map of that paper.

Finally, extension of scalars and suspension give

$$\mathrm{indec}_{P_n} S_1^{-1}(\beta_{n+1})! \simeq \Sigma \mathrm{indec}_{sP_n}(\beta_{n+1})! \simeq \Sigma \mathrm{indec}_{P_{n+1}}.$$

For $n = 0$, the map $P_1 \rightarrow sP_0 = \mathbb{I}$ is its unique reduced augmentation. These identifications are normalized by the usual suspension map on a free generator. $\square$

5. ENHANCED BAR IN POSITIVE SYMMETRIC SEQUENCES

Let $\mathcal{B}_n$ denote the ordinary enhanced n -fold bar, retaining its coaugmented $\mathbb{E}_n$ -coalgebra structure. On the positive objects considered here, this enhanced functor is an equivalence.

Proposition 5.1. *For every finite n , the enhanced bar construction is a right-linear equivalence*

$$(5.1) \quad \mathcal{B}_n : \mathrm{Alg}_{\mathbb{E}_n}^{\mathrm{aug}}(\mathrm{SSeq})_{>0} \xrightarrow{\simeq} \mathrm{Coalg}_{\mathbb{E}_n}^{\mathrm{coaug}}(\mathrm{SSeq})_{>0}.$$

It is symmetric monoidal for the augmented tensor products. If v_n restricts ordinary coalgebra structure along ι_n , there are coherent right-linear equivalences

$$(5.2) \quad v_n \mathcal{B}_{n+1} \simeq \mathcal{B}_n B_{n+1}.$$

The proof first identifies ordinary coalgebras by their cofree comonad. In the associative case the operadic bar theorem then makes indecomposables comonadic, and the bar of a square-zero algebra identifies the ordinary enhanced bar with that comparison functor. Its enhanced symmetric monoidality permits Dunn additivity to recover the higher n and their transition maps.

Lemma 5.2. *For every finite n and r , the spectrum $P_n(r)$ is finite.*

Proof. The case $n = 0$ is immediate. For $n \geq 1$, the map taking little disks to their centers is a homotopy equivalence to the ordered configuration space of r points in $\mathbb{R}^n$. Indeed, over fixed centers the admissible radii form a convex nonempty set. Choosing the radii to be a sufficiently small continuous function of the minimum pairwise distance and the distance to the boundary contracts these fibers.

For $r > 1$, the configuration space is the complement in $\mathbb{R}^{nr}$ of the diagonal subspaces $x_i = x_j$. The coordinate hyperplanes and the hyperplanes $x_{i,k} = x_{j,k}$ give a finite spherical polyhedral decomposition. After subdivision this is a finite triangulation in which the union of diagonals is a subcomplex. Radial retraction reduces its complement to the corresponding complement on the sphere, and subdivision retracts that complement onto the finite dual subcomplex. Thus it has finite CW homotopy type. The case $r = 1$ is contractible. Adding a disjoint basepoint and taking the suspension spectrum proves the assertion. $\square$

Lemma 5.3 (Cofree ordinary coalgebras). *Let $P = P_n$ for $n < \infty$, or let $P = \mathrm{Com}$, with its specified unital little-disks or commutative extension. Over $\mathrm{SSeq}_{>0}$, its ordinary counital, coaugmented coalgebras with positive coaugmentation ideal are the coalgebras for the comonad $P^\vee \circ -$. In particular, their forgetful functor u has right adjoint H , with*

$$(5.3) \quad uH(X) \simeq P^\vee \circ X.$$

For a map $P \rightarrow Q$ compatible with the unital extensions, restriction of ordinary coalgebra structure induces the comonad map $Q^\vee \circ - \rightarrow P^\vee \circ -$. These identifications are right-linear.

Proof. Split off the coaugmentation. By aritywise dualizability, a reduced r -ary operation on its ideal X has the form

$$X \longrightarrow (P(r)^\vee \otimes X^{\otimes D^r})^{h\Sigma_r}.$$

At a nonempty finite set J , only $1 \leq r \leq |J|$ contribute, and

$$(X^{\otimes D^r})(J) = \bigoplus_{\substack{J=J_1 \amalg \dots \amalg J_r \\ J_i \neq \emptyset}} X(J_1) \otimes \dots \otimes X(J_r).$$

The action of Σ_r on ordered partitions is free. Each orbit of summands is therefore an induced Σ_r -object, even after tensoring with the representation $P(r)^\vee$. The norm on this object is an equivalence. Consequently the cofree expression is

$$\begin{aligned} \prod_{r=1}^{|J|} (P(r)^\vee \otimes (X^{\otimes D^r})(J))^{h\Sigma_r} \\ \simeq \bigoplus_{r=1}^{|J|} (P(r)^\vee \otimes (X^{\otimes D^r})(J))_{h\Sigma_r} = (P^\vee \circ X)(J). \end{aligned}$$

To identify the coalgebra structures, use the treewise description of ordinary and divided-power coalgebras and their norm comparison in [4, Appendix A]. After splitting the counit and coaugmentation, all vertices have positive arity, and nonidentity vertices have arity at least two. Over the output set J there are at most $|J| - 1$ nonidentity vertices. At each vertex, the ordered-partition action just described is induced. The norm equivalences are compatible with refinement of partitions: for successive refinements, finite induction and coinduction identify both composites with the same indexed sum. Thus they identify each treewise ordinary operation, and every edge-contraction and refinement map, with the corresponding coaction for $P^\vee \circ -$. This gives inverse constructions on the full tree diagrams, including their mapping spaces. Restoring the coaugmentation supplies the counit terms.

For the cofree property, a map $f : uC \rightarrow X$ extends to $C \rightarrow HX$ by taking, at a partition into r blocks, the reduced r -fold coaction of C followed by $f^{\otimes D^r}$. There are finitely many components at each J . The tree identities say that this is a coalgebra map. Projection onto the unary component recovers f , and the same identities force every coalgebra map to HX to have these components. The construction applies to families parametrized by any space, proving the mapping-space adjunction and (5.3).

Refinement of partitions is the comultiplication of this comonad. Restriction of vertex operations is termwise dual to $P \rightarrow Q$, and replacing a block by a further partition is right substitution. This proves the asserted functoriality. $\square$

Lemma 5.4 (Comonadic lifting). *Let $L : \mathcal{A} \rightleftarrows \mathcal{X} : R$ be comonadic, with counit ϵ , and let S be a comonad on $\mathcal{X}$, with forgetful-cofree adjunction $u : \text{Coalg}_S(\mathcal{X}) \rightleftarrows \mathcal{X} : H$. Suppose $E : \mathcal{A} \rightarrow \text{Coalg}_S(\mathcal{X})$ lifts L . Let*

$$\gamma_X : ERX \longrightarrow HX$$

be the coalgebra map adjoint to $\epsilon_X : LRX \rightarrow X$. If every γ_X is an equivalence, then E is an equivalence.

Proof. Write $T = LR$ and let $\alpha : L \rightarrow SL$ be the coaction of the lift. Its mate is

$$\theta_X : TX \xrightarrow{\alpha_{RX}} STX \xrightarrow{S\epsilon_X} SX.$$

The coaction identities and adjunction triangles make θ a comonad map. For example, naturality at $\eta_A : A \rightarrow RLA$ gives

$$\theta_{LA} L\eta_A = S\epsilon_{LA} \alpha_{RLA} L\eta_A \simeq S(\epsilon_{LA} L\eta_A) \alpha_A \simeq \alpha_A.$$

With $\Delta^T = L\eta R$, naturality at η_{RX} gives

$$\begin{aligned} S\theta_X \theta_{TX} \Delta_X^T &\simeq S^2 \epsilon_X S\alpha_{RX} S(\epsilon_{TX} L\eta_{RX}) \alpha_{RX} \\ &\simeq S^2 \epsilon_X S\alpha_{RX} \alpha_{RX} \simeq \Delta_X^S \theta_X. \end{aligned}$$

The counit identity is similar. These identities are mates of the coherent coaction identities, and the mate construction retains their higher coherence. Under $\mathcal{A} \simeq \text{Coalg}_T(\mathcal{X})$, the functor E is corestriction along θ . Since $\theta_X = u\gamma_X$, the hypothesis makes θ an equivalence of comonads. Hence E is an equivalence. $\square$

Proof of Proposition 5.1. For $n = 1$, apply [4, Theorem 14.18] with right-module operad $\mathbb{I}$ and acting associative operad P_1 . It identifies the enhanced operadic bar from left P_1 -modules to left BP_1 -comodules as an equivalence. Left comodules are coalgebras for $BP_1 \circ -$, so indec_{P_1} is comonadic. The ambient category $\text{SSeq}_{>0}$ is pronilpotent: truncation to arities $\leq d$ is $(d+1)$ -tensor-nilpotent by (4.1), each transition kernel tensors trivially with positive objects, and the inverse limit is $\text{SSeq}_{>0}$.

Take $L = \Sigma \text{indec}_{P_1}$ and $R = \text{triv}_{P_1} \Sigma^{-1}$. By Proposition 4.1, L is the underlying ideal of the ordinary enhanced bar $E = \mathcal{B}_1$. Its target is comonadic by Lemma 5.3. To apply Lemma 5.4, let $V = \Sigma^{-1}X$. The unitized trivial associative algebra is $\mathbf{1}_D \oplus V$, with zero reduced multiplication. Its bar splits as

$$(5.4) \quad u\mathcal{B}_1(\mathbf{1}_D \oplus V) \simeq \bigoplus_{r \geq 1} \Sigma^r V^{\otimes D^r} \simeq \bigoplus_{r \geq 1} X^{\otimes D^r}.$$

Indeed, after splitting the units, the normalized r -simplex is $V^{\otimes D^r}$ with zero reduced face maps. Its realization is $\Sigma^r V^{\otimes D^r}$. The coalgebra maps split an ordered word into consecutive blocks, so (5.4) is the tensor coalgebra with its full deconcatenation structure. It is cofree here: by (4.1), its product and sum expressions agree at every finite set J .

The counit $LRX \rightarrow X$ is projection onto length one, either by the two-sided bar counit or the free-generator normalization in Proposition 4.1. Right linearity also forces its higher-length components to vanish. Thus the map $ERX \rightarrow HX$ in Lemma 5.4 is the cofree identification, proving that $\mathcal{B}_1$ is an equivalence.

To iterate, we show that $\mathcal{B}_1$ is symmetric monoidal. The enhanced bar-cobar constructions are those of [7, Theorem 5.2.2.17, Remark 5.2.2.19, and §5.2.3]. Pass to the retractive category $\text{SSeq}_{\mathbf{1}_D/\mathbf{1}_D}$; its tensor unit is a zero object. Apply [7, Example 5.2.3.11] to the tensor functor on these retractive categories, induced by $\text{SSeq} \times \text{SSeq} \rightarrow \text{SSeq}$. It is symmetric monoidal and preserves geometric realizations jointly, since the diagonal of a sifted indexing category is cofinal. The resulting comparison

$$\mathcal{B}_1(A \otimes_D A') \simeq \mathcal{B}_1(A) \otimes_D \mathcal{B}_1(A')$$

is an equivalence of coalgebras. The same construction for every finite tensor product supplies its unit, symmetry, and associativity coherence. Hence $\mathcal{B}_1$ is a symmetric monoidal equivalence.

Dunn additivity in its unital augmented form [7, Theorem 5.1.2.2] now permits iteration. The interchange

$$\text{Alg}_{\mathbb{E}_a}(\text{Coalg}_{\mathbb{E}_b}(\mathcal{E})) \simeq \text{Coalg}_{\mathbb{E}_b}(\text{Alg}_{\mathbb{E}_a}(\mathcal{E}))$$

uses tensor monoidal structures. Both orders specify the same operations with the same interchange relations: in terms of monoidal envelopes, they interchange the variables indexed by $\text{Env}(\mathbb{E}_a)$ and $\text{Env}(\mathbb{E}_b)^{\text{op}}$. This description also identifies mapping spaces and coherence data. Apply $\mathcal{B}_1$ to one algebra direction at a time, retaining the remaining algebra directions and the coalgebra directions already obtained. Additivity on both sides gives (5.1), agreeing with the enhanced iterated bar of [7, Examples 5.2.3.12 and 5.2.3.14].

Choose the coordinates so that ι_n retains the first n directions and the added direction is processed first. Forgetting its coalgebra structure after the full bar gives exactly (5.2). Both

composites are the same multi-bar diagram read in the two orders; this also gives their compatibility for every composite of adjacent arrows. Finally, R_X is colimit-preserving and symmetric monoidal, so enhanced-bar naturality makes these comparisons right-linear. The inverse of $\mathcal{B}_n$ inherits right linearity from this equivalence. $\square$

6. NORMALIZED COHERENT SELF-DUALITY

Let

$$u_n : \text{Coalg}_{\mathbb{E}_n}^{\text{coaug}}(\text{SSeq})_{>0} \rightleftarrows \text{SSeq}_{>0} : H_n$$

be the forgetful-cofree adjunction of Lemma 5.3. Starting at $n = 0$, (5.2) and (4.3) give identifications

$$(6.1) \quad u_n \mathcal{B}_n \simeq \Sigma^n \text{indec}_{P_n}.$$

We use these identifications recursively, so that the one at $n + 1$ is obtained from the one at n by those two formulas.

Theorem 6.1 (Coherent little-disks self-duality). *There are equivalences of cooperads $h_n : \mathfrak{s}^{-n}BP_n \xrightarrow{\simeq} P_n^\vee$ and equivalences of operads*

$$\phi_n : KP_n \xrightarrow{\simeq} \mathfrak{s}^{-n}P_n$$

with $\phi_0 = \text{id}$, normalized by

$$(6.2) \quad h_n B \phi_n \simeq d_{P_n}.$$

They satisfy

$$(6.3) \quad \phi_n K(\iota_n) \simeq (\mathfrak{s}^{-(n+1)}\beta_{n+1})\phi_{n+1}$$

as an equivalence of diagrams indexed by $\mathbb{N}^{\text{op}}$. Moreover, the enhanced ordinary bar is identified with enhanced operadic bar by

$$(6.4) \quad \mathcal{B}_n(A) \simeq (h_n)_* \mathbf{S}_{-n} \mathbf{B}_{P_n}(A),$$

using Lemma 5.3 and adjoining the coaugmentation on the right.

The normalization in (6.2) will ensure that both enveloping constructions have the same enhanced bar image. We first obtain h_n from the adjunctions underlying (6.1); full faithfulness of reduced operadic bar will then produce the normalized ϕ_n .

Lemma 6.2. *There are equivalences of cooperads*

$$h_n : \mathfrak{s}^{-n}BP_n \xrightarrow{\simeq} P_n^\vee$$

which satisfy, coherently,

$$(6.5) \quad h_n \mathfrak{s}^{-(n+1)} B \beta_{n+1} \simeq \iota_n^\vee h_{n+1},$$

and give (6.4).

Proof. Take right adjoints in (6.1), using Proposition 5.1. This gives

$$H_n \simeq \mathcal{B}_n \text{triv}_{P_n} \Sigma^{-n}$$

and an identification of comonads

$$\begin{aligned} u_n H_n &\simeq \Sigma^n \text{indec}_{P_n} \text{triv}_{P_n} \Sigma^{-n} \\ &\simeq \Sigma^n (BP_n \circ -) \Sigma^{-n} \simeq (\mathfrak{s}^{-n}BP_n) \circ -. \end{aligned}$$

Lemma 5.3 identifies the other side with $P_n^\vee \circ -$. Since this comparison comes from an equivalence of adjunctions, it preserves comultiplication and counit. All its terms and comparisons are right-linear. Thus Lemma 4.2 recovers the cooperad equivalence h_n .

The comparison functor for $\Sigma^n \text{indec}_{P_n} \dashv \text{triv}_{P_n} \Sigma^{-n}$ is $\mathbf{S}_{-n} \mathbf{B}_{P_n}$. The comparison functor for $u_n \dashv H_n$ is the identity of the coalgebra category in Lemma 5.3. The equivalence of adjunctions identifies these comparison functors, proving (6.4) with its coaction.

For the transitions apply the mate formula of Lemma 2.1 to β_{n+1} and transport it through (5.2). On the algebra side the resulting comonad map is $\mathbf{s}^{-(n+1)} B\beta_{n+1}$. On the ordinary-coalgebra side it is the map for restriction along ι_n , namely $\iota_n^\vee$. The recursive choice of (6.1) identifies their units and counits, giving (6.5). Applying mates to the whole diagram supplies the coherence of these transitions. $\square$

Proof of Theorem 6.1. By Lemmas 2.2 and 6.2, the composite

$$BKP_n \xrightarrow{d_{P_n}} P_n^\vee \xrightarrow{h_n^{-1}} \mathbf{s}^{-n} BP_n \simeq B(\mathbf{s}^{-n} P_n)$$

is an equivalence of reduced cooperads. Reduced operadic bar is fully faithful, so this is $B\phi_n$ for an operad equivalence ϕ_n , with a contractible space of lifts including the specified identification. This proves (6.2); at level zero all the maps are identities. Naturality of d and (6.5) give

$$\begin{aligned} h_n B\phi_n BK(\iota_n) &\simeq d_{P_n} BK(\iota_n) \\ &\simeq \iota_n^\vee d_{P_{n+1}} \\ &\simeq \iota_n^\vee h_{n+1} B\phi_{n+1} \\ &\simeq h_n \mathbf{s}^{-(n+1)} B\beta_{n+1} B\phi_{n+1}. \end{aligned}$$

Cancel h_n and use full faithfulness of B to obtain (6.3). Both d and (6.5) are maps of cooperad diagrams; applying an inverse of the reduced bar equivalence lifts this diagram with its higher commutativity data. Formula (6.4) was proved in Lemma 6.2. $\square$

7. COMPARISON OF THE ENVELOPING-ALGEBRA CONES

We first recall the internal loop enhancement used to define the ordinary enveloping functors. The monoidal structure on the algebra category in the next lemma is cartesian.

Lemma 7.1 (Internal delooping). *For any reduced spectral operad P and finite n , internal loops give an equivalence*

$$\tilde{\Omega}_P^n : \text{Alg}_P(\mathcal{C}) \xrightarrow{\sim} \text{Alg}_{\mathbb{E}_n}(\text{Alg}_P(\mathcal{C})).$$

Its inverse is internal n -fold bar, denoted Bar_P^n . Both preserve geometric realizations and are natural under colimit-preserving symmetric monoidal functors between the ambient stable categories.

Proof. This is [8, Proposition 2.2.9 and Remark 2.2.10]. The forgetful functor to $\mathcal{C}$ is conservative, creates finite limits, and preserves geometric realizations. Finite products become finite direct sums in $\mathcal{C}$. Consequently the underlying internal bar is the bar for its cocartesian structure and has value Σ^n of the underlying object; the underlying internal cobar has value Ω^n . The bar-cobar unit and counit become the suspension-loop unit and counit after forgetting, so they are equivalences. The simplicial additive bar gives $n = 1$, and iteration gives finite n .

On underlying stable objects, realizations commute with the finite limits defining loops. Thus the loop enhancement preserves realizations, as does internal bar by construction. A colimit-preserving functor between stable categories is exact and therefore preserves these finite limits and bar realizations. When it is symmetric monoidal, it also preserves their operadic structures, which proves the naturality assertion. $\square$

By Theorem 3.1, $F = \mathbf{1} \oplus \text{indec}_{\text{Lie}}$ is symmetric monoidal from cartesian Lie algebras to augmented objects with their tensor product. Define

$$(7.1) \quad U_n(\mathfrak{g}) = \text{the augmentation ideal of } F(\tilde{\Omega}_{\text{Lie}}^n \mathfrak{g}),$$

where F is applied to the internal $\mathbb{E}_n$ -algebra structure. It preserves the relevant realizations by Lemma 2.1. Internal bar and loops give the coherent comparisons

$$(7.2) \quad B_{n+1}U_{n+1} \simeq U_n, \quad U_0 = \text{indec}_{\text{Lie}}.$$

This is the construction of [8, §2.2.13] in the general ambient category $\mathcal{C}$.

Using the normalized self-duality of Theorem 6.1, put

$$(7.3) \quad c_n = \phi_n K(i_n) : \text{Lie} \longrightarrow \mathfrak{s}^{-n} P_n, \quad \tilde{U}_n = S_{-n}^{-1}(c_n)!$$

Extension along $K(\iota_n)$, transported through ϕ_n and the suspension equivalences, defines the corresponding horizontal functors $\tilde{B}_{n+1}$.

Theorem 7.2 (Comparison of enveloping-algebra cones). *For every presentably symmetric monoidal stable ∞ -category $\mathcal{C}$, there are natural equivalences*

$$U_n \simeq \tilde{U}_n, \quad B_{n+1} \simeq \tilde{B}_{n+1}$$

which identify the entire cones in

$$\text{Fun}((\mathbb{N}^{\text{op}})^{\triangleleft}, \text{Pr}^{\text{L}}).$$

The same conclusion, as an equivalence of cones, holds for any choice of little-disks self-duality equivalences, after conjugating the vertices by their induced operadic automorphisms.

The two enveloping functors will first be compared on positive symmetric sequences. Their enhanced bars are both the restriction of the same Chevalley–Eilenberg coalgebra; the normalization (6.2) is what identifies these coalgebra structures. The equivalence (5.1) then compares the functors and their cones there. Finally their right-linear kernels carry this comparison, including its substitution maps, to arbitrary $\mathcal{C}$.

For the next two propositions work in $\mathbf{SSeq}$, with input in $\text{Alg}_{\text{Lie}}(\mathbf{SSeq}_{>0})$. The constructions preserve positivity. Let

$$r_n : \text{Coalg}_{\mathbb{E}_\infty}^{\text{coaug}}(\mathbf{SSeq})_{>0} \longrightarrow \text{Coalg}_{\mathbb{E}_n}^{\text{coaug}}(\mathbf{SSeq})_{>0}$$

restrict structure along i_n .

Proposition 7.3. *There are natural right-linear equivalences*

$$(7.4) \quad B_n U_n \simeq r_n \text{CE}.$$

They identify the enhanced bar of the cone (7.2) with the cone of restrictions of the single cocommutative coalgebra CE.

Proof. Apply enhanced-bar naturality [7, Example 5.2.3.11] to the symmetric monoidal, realization-preserving functor F . Its source is the cartesian monoidal category of Lie algebras, with zero algebra as unit. Its target is the category of augmented objects of $\mathbf{SSeq}$, whose unit is also a zero object in that augmented category. Enhanced-bar naturality gives an equivalence of coalgebras

$$B_n U_n(\mathfrak{g}) \simeq F(\text{Bar}_{\text{Lie}}^n \tilde{\Omega}_{\text{Lie}}^n \mathfrak{g}) \simeq F(\mathfrak{g})$$

by Lemma 7.1.

In a cartesian symmetric monoidal category, the counital coalgebra structure is the canonical diagonal, with its contractible space of coherent choices. The monoidal structure on F was induced by the product-preserving functor CE. It therefore takes this diagonal to the existing coproduct on $\text{CE}(\mathfrak{g})$: its binary map is the image of $\mathfrak{g} \rightarrow \mathfrak{g} \times \mathfrak{g}$ followed by the canonical product comparison, and the same holds in every arity. This identifies the coalgebra structure with $r_n \text{CE}$.

Apply the same naturality to the multi-bar diagrams. Their transition maps become restriction of this one coproduct, giving the claimed comparison of cones. Right substitution commutes with F , its canonical monoidal structure, and internal loops by Lemmas 2.1, 4.2, and 7.1. The comparison is consequently right-linear. $\square$

Proposition 7.4. *There are natural right-linear equivalences*

$$(7.5) \quad \mathcal{B}_n \tilde{U}_n \simeq r_n \text{ CE}$$

which respect the entire Koszul-dual cone.

Proof. Use (6.4), suspension naturality, and Lemma 2.1. The resulting $BLie$ -coalgebra is $\mathbf{B}_{Lie}(\mathfrak{g})$, with its coaction corestricted along

$$(7.6) \quad BLie \xrightarrow{BK(i_n)} BK P_n \xrightarrow{B\phi_n} \mathfrak{s}^{-n} B P_n \xrightarrow{h_n} P_n^\vee.$$

Equation (6.2) and the naturality of d in Lemma 2.2 identify this map with

$$(7.7) \quad BLie \xrightarrow{d_{Com}} Com \xrightarrow{i_n^\vee} P_n^\vee.$$

In positive symmetric sequences, Lemma 5.3 identifies ordinary coalgebras and divided-power comonad coalgebras through the specified norm equivalences. Under this identification, (7.7) is exactly the restriction of the ordinary Chevalley–Eilenberg structure. This proves (7.5) as a coalgebra equivalence.

All maps in (7.6) and (7.7) belong to the coherent operad and cooperad diagrams already constructed. They respect $i_{n+1}\iota_n = i_n$ and (6.3). Thus they identify the entire cone with the restrictions of one CE object. The operations and comparisons are right-linear, giving the asserted linearity as well. $\square$

Corollary 7.5. *On $S\text{Seq}_{>0}$, the functors U_n and $\tilde{U}_n$ are equivalent as right-linear P_n -algebra-valued functors, and their entire cones are equivalent.*

Proof. Propositions 7.3 and 7.4 identify their images under $\mathcal{B}_n$, with coalgebra structures and all cone maps. Apply its inverse in the right-module functor category using Proposition 5.1.

For the horizontal functors, let

$$t_n = \phi_n K(\iota_n) \phi_{n+1}^{-1} \simeq \mathfrak{s}^{-(n+1)} \beta_{n+1}.$$

Then

$$\tilde{B}_{n+1} = S_{-n}^{-1}(t_n)! S_{-(n+1)} \simeq S_1^{-1}(\beta_{n+1})! \simeq B_{n+1}.$$

The underlying suspension in the last expression is Σ . Both enhanced cones have been identified with

$$\cdots \longrightarrow r_{n+1} \text{ CE} \longrightarrow r_n \text{ CE} \longrightarrow \cdots \longrightarrow r_0 \text{ CE}.$$

This gives the triangles at the Lie vertex as well as the horizontal comparisons. $\square$

Lemma 7.6 (Evaluation and kernels). *For $X \in \mathcal{C}$, the functor*

$$(7.8) \quad \text{ev}_X : S\text{Seq} \longrightarrow \mathcal{C}, \quad Y \longmapsto \bigoplus_{r \geq 0} (Y(r) \otimes X^{\otimes r})_{h\Sigma_r}$$

is colimit-preserving, exact, and symmetric monoidal for Day convolution, and sends $\mathbb{I}$ to X .

Suppose a family $T_{\mathcal{C}} : \text{Alg}_P(\mathcal{C}) \rightarrow \text{Alg}_Q(\mathcal{C})$ preserves geometric realizations and commutes with these evaluation functors by comparisons natural in X and compatible with right substitution. If its restriction to positive symmetric sequences is right-linear, let $N_T = T_{S\text{Seq}_{>0}}(P)$ as in Lemma 4.3. Then

$$(7.9) \quad T_{\mathcal{C}}(A) \simeq N_T \circ_P A.$$

Right-linear comparisons of such families, and coherent diagrams of comparisons, extend accordingly to every $\mathcal{C}$.

Proof. Formula (7.8) is the Day extension of the assignment of X to a one-element set. Reindexing finite partitions proves its symmetric monoidality, as in Lemma 4.2. Tensor products and orbits preserve colimits, so evaluation does too; exactness follows from stability.

Since $\text{ev}_X(P) = \text{Free}_P X$, compatibility with evaluation gives

$$T_{\mathcal{C}}(\text{Free}_P X) \simeq \text{ev}_X(N_T) = N_T \circ X.$$

For a map $f : \text{Free}_P X \rightarrow \text{Free}_P Y$ adjoint to $\bar{f} : X \rightarrow P \circ Y$, its canonical factorization is

$$(7.10) \quad \text{Free}_P X \xrightarrow{\text{Free}_P(\bar{f})} \text{Free}_P(P \circ Y) \xrightarrow{\mu_{P,Y}} \text{Free}_P Y.$$

Naturality in X identifies the first arrow after applying T with $N_T \circ \bar{f}$. Compatibility with right substitution identifies the second with the right P -action $N_T \circ P \circ Y \rightarrow N_T \circ Y$ from Lemma 4.3. Thus the displayed free-object comparison respects arbitrary maps between free algebras, including the maps in their simplicial resolutions.

Apply T to the standard simplicial free resolution of A . By (7.10), its image is $B_{\bullet}(N_T, P, A)$ with the prescribed faces and degeneracies. Realization proves (7.9), retaining the left Q -structure. Evaluation, free resolution, and realization are functorial constructions; the same argument therefore applies to natural transformations and arbitrary indexing diagrams. $\square$

Proof of Theorem 7.2. The Koszul-dual functors are made from suspension and extension of scalars and thus commute with (7.8). The ordinary enveloping functors use enhanced CE, its canonical monoidal structure, and internal loops. Enhanced CE commutes with evaluation by Lemma 2.1. Its monoidal structure does too: it is inverse to the canonical product comparison of Theorem 3.1, and exact symmetric monoidal evaluation preserves that comparison and finite products of Lie algebras. Exactness and Lemma 7.1 likewise make evaluation commute with the loop enhancement. Enhanced-bar naturality does the same for ordinary bar. These comparisons are natural in the evaluating object and compatible with right substitution, since the constructions use functorial finite limits, tensor products, norms, and realizations. They retain the indicated algebra and coalgebra structures.

All these functors preserve geometric realizations. For U_n , finite limits defining Lie loops commute with realizations after forgetting to the ambient stable category, and $\text{indec}_{\text{Lie}}$ preserves realizations. The relevant forgetful functors create them. Extension of scalars and ordinary bar have the same property directly from their bar formulas.

Corollary 7.5 therefore gives an equivalence of bimodules

$$N_{U_n} \simeq N_{\tilde{U}_n} \quad \text{in} \quad \text{Bimod}_{(P_n, \text{Lie})}(\text{SSeq}_{>0}).$$

Lemma 7.6 now gives, for every $A \in \text{Alg}_{\text{Lie}}(\mathcal{C})$,

$$U_n(A) \simeq N_{U_n} \circ_{\text{Lie}} A \simeq N_{\tilde{U}_n} \circ_{\text{Lie}} A \simeq \tilde{U}_n(A).$$

The right-linear transformations for the horizontal arrows and the cone triangles give the corresponding maps of kernels, whose composition is relative composition product. Functorial realization transports this cone comparison to $\mathcal{C}$. The Koszul-dual functors are left adjoints; the equivalences show the same for U_n and B_{n+1} . Hence the comparison takes place in the stated functor category with values in Pr^{L} .

Finally let $\psi_n : KP_n \xrightarrow{\simeq} \mathfrak{s}^{-n}P_n$ be another self-duality choice and set

$$a_n = \mathfrak{s}^n(\phi_n \psi_n^{-1}) : P_n \xrightarrow{\simeq} P_n.$$

Extension along a_n is an autoequivalence of $\text{Alg}_{P_n}(\mathcal{C})$, and substitution in the definitions gives

$$\tilde{U}_n^{\phi} \simeq (a_n)_! \tilde{U}_n^{\psi}, \quad \tilde{B}_{n+1}^{\phi} \simeq (a_n)_! \tilde{B}_{n+1}^{\psi} (a_{n+1})_!^{-1}.$$

These are the coherent conjugacy equivalences of the transported Koszul-dual diagram and retain the Lie vertex. Thus arbitrary choices give equivalent cones by conjugation, as required in [8, Conjecture 2.4.10]. $\square$

Theorem 1.1 follows from Theorem 3.1, the cone construction (7.2), and Theorem 7.2.

8. THE RATIONAL EQUIVALENCE

Theorem 7.2 identifies the rational Koszul-dual cone with the actual-bar cone. Shi's rational equivalence therefore gives the following result.

Theorem 8.1. *The actual higher enveloping functor is an equivalence*

$$U_\infty: \text{Alg}_{\text{Lie}}(\text{Sp}_{\mathbb{Q}}) \xrightarrow{\simeq} \text{BarTower}_0.$$

Proof. By [8, Theorem 2.4.12], the functor from $\text{Alg}_{\text{Lie}}(\text{Sp}_{\mathbb{Q}})$ to the inverse limit of the Koszul-dual enveloping-algebra tower is an equivalence. The category $\text{Sp}_{\mathbb{Q}}$ is presentably symmetric monoidal and stable, so theorem 7.2 applies. It gives an equivalence of the entire Koszul-dual cone with the actual-bar cone, including the Lie vertex. Taking inverse limits identifies the induced functor with U_∞ and proves the assertion. $\square$

This proves the height-zero part of theorem 1.2(i). Appendix A gives a direct proof using actual bar: it identifies the suspension of a rational tower as a free tower in proposition A.3 and computes the adjunction unit on trivial Lie algebras.

9. BAR TOWERS AT POSITIVE HEIGHTS

Fix $h \geq 1$. The categories $\text{Alg}_{\mathbb{E}_n}^{\text{nu}}(\text{Sp}_{T(h)})$ and BarTower_h are pointed and presentable. Colimits in BarTower_h are computed levelwise.

We use the equivalence between nonunital and augmented unital $\mathbb{E}_n$ -algebras in a stable symmetric monoidal category, given by adjoining a unit and taking its augmentation ideal [7, Proposition 5.4.4.10]. Under this equivalence, reduced bar corresponds to augmented bar. Write

$$i_n!: \text{Alg}_{\mathbb{E}_n}^{\text{nu}}(\text{Sp}_{T(h)}) \rightleftarrows \text{Alg}_{\mathbb{E}_{n+1}}^{\text{nu}}(\text{Sp}_{T(h)}) : i_n^*$$

for extension and restriction along $\mathbb{E}_n \rightarrow \mathbb{E}_{n+1}$.

Proposition 9.1. *The loop functor of BarTower_h is given by*

$$(9.1) \quad (\Omega_{\text{BarTower}_h} A)_n \simeq i_n^* A_{n+1}.$$

In particular, $\Omega_{\text{BarTower}_h}$ is conservative.

The proof uses the bar-suspension comparison and an adjunction on the inverse limit. The comparison of Heuts-Land gives

$$(9.2) \quad i_n! B_{n+1} \simeq \Sigma_{\text{Alg}_{\mathbb{E}_{n+1}}^{\text{nu}}(\text{Sp}_{T(h)})}.$$

This is the lower-left triangle of [5, Theorem 2.4], with their index equal to $n+1$; in particular, it applies also when $n=0$.

Lemma 9.2. *Consider two towers $(\mathcal{X}_n, u_{n+1})$ and $(\mathcal{Y}_n, v_{n+1})$ and adjunctions $L_n: \mathcal{X}_n \rightleftarrows \mathcal{Y}_n: G_n$. Suppose there are coherent equivalences*

$$L_n u_{n+1} \simeq v_{n+1} L_{n+1}$$

whose mates $u_{n+1} G_{n+1} \rightarrow G_n v_{n+1}$ are equivalences. Then the levelwise functors induce an adjunction on the inverse limits.

Proof. The comparisons define functors L and G on compatible towers. For compatible objects X and Y , the mapping space in an inverse limit of categories is the inverse limit of the corresponding mapping spaces. The levelwise adjunction equivalences

$$\mathrm{Map}_{Y_n}(L_n X_n, Y_n) \simeq \mathrm{Map}_{X_n}(X_n, G_n Y_n)$$

commute with the transition maps because the two comparisons are mates. Taking their limit gives an equivalence $\mathrm{Map}(LX, Y) \simeq \mathrm{Map}(X, GY)$, natural in both variables. $\square$

Proof of proposition 9.1. The canonical comparisons

$$(9.3) \quad B_{n+1} i_{n+1}^* \longrightarrow i_n^* B_{n+2}$$

are equivalences. In augmented algebras, both sides are the realization of the same two-sided simplicial bar after restricting the residual operadic structure: restriction preserves tensor products and geometric realizations. The mate identification is given explicitly in [2, Lemma 3.3 and proof of Theorem 1.6]. This argument applies in $\mathrm{Sp}_{T(h)}$, since its tensor product preserves colimits in each variable.

Apply lemma 9.2 to the adjunctions $i_n! \dashv i_n^*$ and the tower with its first term omitted. The left comparison is the commuting bar-extension square of [5, Theorem 2.4], and its right mate is (9.3). Thus the right adjoint has the terms in (9.1). For a compatible tower X , the corresponding left adjoint has terms

$$i_n! X_n \simeq i_n! B_{n+1} X_{n+1} \simeq \Sigma_{\mathrm{Alg}_{\mathbb{E}_{n+1}}^{\mathrm{nu}}(\mathrm{Sp}_{T(h)})} X_{n+1}.$$

These equivalences respect the tower structure: the bar-extension square is obtained from the same suspension triangles used in (9.2). Dropping the zeroth term is an equivalence of the two inverse-limit categories. Since suspension is computed levelwise, this left adjoint is $\Sigma_{\mathrm{BarTower}_h}$, proving (9.1).

If $\Omega_{\mathrm{BarTower}_h} f$ is an equivalence, conservativity of each restriction functor implies that f_n is an equivalence for $n \geq 1$. Applying B_1 gives the same conclusion at $n = 0$. Hence f is an equivalence. $\square$

For a pointed category $\mathcal{X}$ with finite colimits, its costabilization is

$$\mathrm{coSp}(\mathcal{X}) = \varprojlim (\cdots \xrightarrow{\Sigma_{\mathcal{X}}} \mathcal{X} \xrightarrow{\Sigma_{\mathcal{X}}} \mathcal{X}).$$

It is stable. Evaluation at zero, denoted by ev_0 , is universal for finite-colimit-preserving functors from stable categories [8, Propositions 3.1.4, 3.1.7, and 3.1.9].

Proposition 9.3. *There is an equivalence $\mathrm{Sp}_{T(h)} \simeq \mathrm{coSp}(\mathrm{BarTower}_h)$ under which evaluation at zero is $U_\infty \mathrm{Free}_{\mathrm{Lie}}$. Every object $Z \in \mathrm{BarTower}_h$ admitting an equivalence $\Sigma_{\mathrm{BarTower}_h}^d Z \simeq Z$ for some $d > 0$ belongs to $\mathrm{EssIm}(U_\infty \mathrm{Free}_{\mathrm{Lie}})$.*

Proof. The proof of [8, Proposition 3.3.3] establishes that

$$\mathrm{coSp}(\mathrm{pr}_0): \mathrm{coSp}(\mathrm{BarTower}_h) \longrightarrow \mathrm{Sp}_{T(h)}$$

is an equivalence. The costabilization universal property lifts $U_\infty \mathrm{Free}_{\mathrm{Lie}}$ to $\mathrm{coSp}(\mathrm{BarTower}_h)$. Since

$$\mathrm{pr}_0 U_\infty \mathrm{Free}_{\mathrm{Lie}} \simeq \mathrm{id}_{\mathrm{Sp}_{T(h)}}$$

by [8, Proposition 2.2.14, at $n = 0$], this lift is inverse to $\mathrm{coSp}(\mathrm{pr}_0)$. Its evaluation at zero is $U_\infty \mathrm{Free}_{\mathrm{Lie}}$, as asserted.

Passing to the cofinal subsequence of indices divisible by d identifies $\mathrm{coSp}(\mathrm{BarTower}_h)$ with the inverse limit along $\Sigma_{\mathrm{BarTower}_h}^d$. The given periodicity equivalence defines an object of this limit with value Z at zero. The first assertion now applies. $\square$

10. MAPPING SPACES AND THE HEIGHT-ONE CASE

For a pointed space K and $X \in \text{BarTower}_h$, let $K \odot X$ denote the reduced tensor of X with K . Its defining adjunction is

$$(10.1) \quad \text{Map}_{\text{BarTower}_h}(K \odot X, A) \simeq \text{Map}_*(K, \text{Map}_{\text{BarTower}_h}(X, A)).$$

The basepoint of the mapping space is the zero morphism.

Lemma 10.1. *If a map of pointed spaces $f: K \rightarrow L$ induces an equivalence $L_{T(h)}\Sigma^\infty f$, then $f \odot X$ is an equivalence for every $X \in \text{BarTower}_h$. Consequently, $\text{Map}_{\text{BarTower}_h}(X, A)$ is local with respect to all such maps f .*

Proof. First work in $\text{Alg}_{\mathbb{E}_n}^{\text{nu}}(\text{Sp}_{T(h)})$. The forgetful functor creates limits, so the reduced cotensor A^K has underlying spectrum

$$\text{oblv}(A^K) \simeq \text{map}(\Sigma^\infty K, \text{oblv } A).$$

Since $\text{oblv } A$ is $T(h)$ -local, the map $A^L \rightarrow A^K$ is an equivalence. Taking left adjoints gives the assertion for reduced tensors in $\text{Alg}_{\mathbb{E}_n}^{\text{nu}}(\text{Sp}_{T(h)})$. Reduced tensors in BarTower_h are computed levelwise because the bar functors preserve colimits. The last assertion follows from (10.1). $\square$

Lemma 10.2. *Every mapping space $\text{Map}_{\text{BarTower}_h}(X, A)$ is local for mod- p homology equivalences of pointed spaces.*

Proof. It suffices, by lemma 10.1, to prove that every mod- p homology equivalence of spaces induces a $T(h)$ -equivalence on suspension spectra. Its spectral cofiber Z is bounded below and satisfies $H\mathbb{F}_p \wedge Z = 0$. Put $M = S/p \wedge Z$. If $M \neq 0$, let k be the least integer with $\pi_k M \neq 0$. The Moore cofiber sequence gives an exact sequence

$$0 \longrightarrow \pi_k Z/p \longrightarrow \pi_k M \longrightarrow \pi_{k-1} Z[p] \longrightarrow 0.$$

Thus $p^2 \pi_k M = 0$. A nonzero group killed by p^2 has a nonzero quotient modulo p . The lowest-degree stable Hurewicz theorem therefore gives

$$H_k(M; \mathbb{F}_p) \cong \pi_k M/p \neq 0,$$

contradicting $H\mathbb{F}_p \wedge M = 0$. Hence $M = 0$, so multiplication by p is invertible on Z and $Z \simeq Z[1/p]$.

Since $h > 0$, the finite type- h spectrum defining $T(h)$ is rationally acyclic. It follows that $T(h)[1/p] = 0$. Consequently $T(h) \wedge Z = 0$, as required. $\square$

Lemma 10.3. *Let V be a finite pointed space equipped with a map $v: \Sigma^d V \rightarrow V$, where $d > 0$, whose suspension spectrum is a $T(h)$ -equivalence. If $\text{oblv}_{\text{Lie}} T_\infty(A) = 0$, then*

$$\text{Map}_*(V, \text{Map}_{\text{BarTower}_h}(X, A)) \simeq * \quad \text{for every } X \in \text{BarTower}_h.$$

Proof. By lemma 10.1, the map

$$\Sigma_{\text{BarTower}_h}^d(V \odot X) \simeq (\Sigma^d V) \odot X \xrightarrow{v \odot X} V \odot X$$

is an equivalence. Proposition 9.3 implies $V \odot X \simeq U_\infty \text{Free}_{\text{Lie}}(E)$ for some $E \in \text{Sp}_{T(h)}$. Now (10.1) and $U_\infty \text{Free}_{\text{Lie}} \dashv \text{oblv}_{\text{Lie}} T_\infty$ identify the asserted mapping space with $\text{Map}_{\text{Sp}_{T(h)}}(E, \text{oblv}_{\text{Lie}} T_\infty(A))$, which is contractible. $\square$

Lemma 10.4. *Let $a \geq 2$, and let Y be a pointed space local for mod- p homology equivalences. If*

$$\text{Map}_*(M(\mathbb{Z}/p, a), Y) \simeq *, \quad M(\mathbb{Z}/p, a) = \text{cofib}(p: S^a \rightarrow S^a),$$

*then $\Omega^{a+1}Y \simeq *$.*

Proof. Applying $\text{Map}_*(-, Y)$ to the Moore cofiber sequence shows that the homotopy fiber over the basepoint of

$$p: \Omega^a Y \longrightarrow \Omega^a Y$$

is contractible. Its long exact homotopy sequence implies that multiplication by p is invertible on every positive homotopy group of $\Omega^a Y$.

Let $W = (\Omega^a Y)_0$. The p th-power map of this connected loop space is a homotopy equivalence, since it induces multiplication by p on all positive homotopy groups. We claim that $\tilde{H}_*(W; \mathbb{F}_p) = 0$. Otherwise choose the least positive degree m with nonzero reduced homology. By the Künneth theorem, the diagonal of a class in $H_m(W; \mathbb{F}_p)$ is primitive: all terms involving positive homology in two factors vanish by minimality. The p th-power map therefore acts on H_m as multiplication by p , hence as zero. This contradicts its being a homotopy equivalence.

Local spaces are closed under limits. Moreover, every discrete space is mod- p -homology local, since a mod- p homology equivalence induces a bijection on components. It follows that

$$W \simeq \text{fib}(\Omega^a Y \longrightarrow \pi_0(\Omega^a Y))$$

is local. Thus $W \rightarrow *$ is both a mod- p homology equivalence and a map between local spaces, so it is an equivalence. Finally, $\Omega^{a+1} Y \simeq \Omega W \simeq *$. $\square$

Proof of the height-one case of theorem 1.2(i). Set $h = 1$. We first prove that $\text{oblv}_{\text{Lie}} T_\infty(A) = 0$ forces $A = 0$, using a uniform Moore-nullity bound on mapping spaces. The fiber argument and conservativity of loops then show that $\text{oblv}_{\text{Lie}} T_\infty$ detects equivalences. Applying this to the counit of $U_\infty \dashv T_\infty$ proves the theorem.

The mod- p Moore spectrum is finite of type one. By the periodicity theorem [6, Theorem 9], it admits a positive-degree v_1 self-map. A stable map between finite suspension spectra is represented by a map of spaces after sufficiently many suspensions. Thus, for some $a \geq 2$, there is an actual map

$$v: \Sigma^d M(\mathbb{Z}/p, a) \longrightarrow M(\mathbb{Z}/p, a)$$

whose suspension spectrum is a $T(1)$ -equivalence. Here a and d are fixed, independently of the objects of BarTower_h . This argument applies at $p = 2$ as well, after choosing an appropriate power of the periodic self-map.

Suppose $\text{oblv}_{\text{Lie}} T_\infty(A) = 0$. By lemmas 10.2 to 10.4, for every $X \in \text{BarTower}_h$ we have

$$\text{Map}_{\text{BarTower}_h}(X, \Omega_{\text{BarTower}_h}^{a+1} A) \simeq \Omega^{a+1} \text{Map}_{\text{BarTower}_h}(X, A) \simeq *.$$

Yoneda gives $\Omega_{\text{BarTower}_h}^{a+1} A = 0$, and proposition 9.1 then gives $A = 0$.

We next show that $\text{oblv}_{\text{Lie}} T_\infty$ detects equivalences. If $\text{oblv}_{\text{Lie}} T_\infty(f)$ is an equivalence, the fiber P of $f: A \rightarrow B$ satisfies $\text{oblv}_{\text{Lie}} T_\infty(P) = 0$, since $\text{oblv}_{\text{Lie}} T_\infty$ is a right adjoint. Hence $P = 0$. In any pointed category with finite limits, the canonical pullback identifications give

$$\Omega B = 0 \times_B 0 \simeq 0 \times_A (A \times_B 0) \simeq 0 \times_A 0 = \Omega A.$$

Thus Ωf is an equivalence. Conservativity of $\Omega_{\text{BarTower}_h}$ implies that f is an equivalence.

Finally, full faithfulness of U_∞ and the triangle identities imply that the counit

$$U_\infty T_\infty A \longrightarrow A$$

becomes an equivalence under T_∞ , and therefore under $\text{oblv}_{\text{Lie}} T_\infty$. The conservativity just proved makes the counit an equivalence. This proves essential surjectivity and completes the height-one case. $\square$

11. COMMUTATIVE COSPECTRA AND BAR TOWERS

Continue to fix $h \geq 1$. Let

$$\text{com}_n : \text{Alg}_{\mathbb{E}_n}^{\text{nu}}(\text{Sp}_{T(h)}) \rightleftarrows \text{CAlg}^{\text{nu}}(\text{Sp}_{T(h)}) : \text{res}_{\mathbb{E}_n}^{\mathbb{E}_\infty}$$

be commutativization and restriction. An object $C_\bullet \in \text{coSp}(\text{CAlg}^{\text{nu}}(\text{Sp}_{T(h)}))$ is a sequence with equivalences $\Sigma_{\text{CAlg}^{\text{nu}}(\text{Sp}_{T(h)})} C_{n+1} \simeq C_n$.

Proposition 11.1. *Levelwise commutativization and restriction define an adjunction*

$$\text{com}_h^{\text{bar}} : \text{BarTower}_h \rightleftarrows \text{coSp}(\text{CAlg}^{\text{nu}}(\text{Sp}_{T(h)})) : \text{res}_h^{\text{bar}},$$

whose terms are

$$(\text{com}_h^{\text{bar}} A)_n = \text{com}_n A_n, \quad (\text{res}_h^{\text{bar}} C_\bullet)_n = \text{res}_{\mathbb{E}_n}^{\mathbb{E}_\infty} C_n.$$

Proof. Transitivity of operadic extension gives $\text{com}_n \simeq \text{com}_{n+1} i_{n!}$. Since com_{n+1} preserves colimits, (9.2) gives the comparison

$$(11.1) \quad \text{com}_n B_{n+1} \simeq \text{com}_{n+1} i_{n!} B_{n+1} \simeq \text{com}_{n+1} \Sigma_{\text{Alg}_{\mathbb{E}_{n+1}}^{\text{nu}}}(\text{Sp}_{T(h)}) \simeq \Sigma_{\text{CAlg}^{\text{nu}}(\text{Sp}_{T(h)})} \text{com}_{n+1}.$$

To apply lemma 9.2, we show that its mate

$$(11.2) \quad B_{n+1} \text{res}_{\mathbb{E}_{n+1}}^{\mathbb{E}_\infty} \longrightarrow \text{res}_{\mathbb{E}_n}^{\mathbb{E}_\infty} \Sigma_{\text{CAlg}^{\text{nu}}(\text{Sp}_{T(h)})}$$

is an equivalence.

Let C be a nonunital commutative algebra and $C^+ = \mathbf{1} \oplus C$ its corresponding augmented algebra. The suspension of C^+ in augmented commutative algebras is the pushout $\mathbf{1} \amalg_{C^+} \mathbf{1}$. This is the relative tensor product $\mathbf{1} \otimes_{C^+} \mathbf{1}$, calculated by the simplicial bar object

$$[q] \longmapsto \mathbf{1} \otimes (C^+)^{\otimes q} \otimes \mathbf{1}.$$

Restriction to $\mathbb{E}_n$ -algebras preserves the tensor product and geometric realizations. The augmentation ideal of the restricted realization is therefore $B_{n+1} \text{res}_{\mathbb{E}_{n+1}}^{\mathbb{E}_\infty} C$, with its residual $\mathbb{E}_n$ -structure.

The mate is the composite

$$\begin{aligned} B_{n+1} \text{res}_{\mathbb{E}_{n+1}}^{\mathbb{E}_\infty} C &\longrightarrow \text{res}_{\mathbb{E}_n}^{\mathbb{E}_\infty} \text{com}_n B_{n+1} \text{res}_{\mathbb{E}_{n+1}}^{\mathbb{E}_\infty} C \\ &\longrightarrow \text{res}_{\mathbb{E}_n}^{\mathbb{E}_\infty} \Sigma_{\text{CAlg}^{\text{nu}}(\text{Sp}_{T(h)})} \text{com}_{n+1} \text{res}_{\mathbb{E}_{n+1}}^{\mathbb{E}_\infty} C \longrightarrow \text{res}_{\mathbb{E}_n}^{\mathbb{E}_\infty} \Sigma_{\text{CAlg}^{\text{nu}}(\text{Sp}_{T(h)})} C. \end{aligned}$$

The first and last arrows come from the unit and counit. The comparison in [5, proof of Theorem 2.4, using Lemma 2.3] is constructed from the canonical map from bar to the suspension pushout. After the displayed unit and counit, it is precisely the map on augmentation ideals induced by the identification of the two bar realizations just described. That map is an equivalence. These constructions are natural and use the same simplicial bar, so supply the coherent comparisons required by lemma 9.2. The proposition follows. $\square$

The free nonunital commutative-algebra functor $\text{Free}_{\text{Com}}^{\text{nu}}$ has a finite-colimit-preserving lift to costabilization. Denote this lift and its right adjoint by

$$\text{Free}_{\text{Com}}^{\text{co}} : \text{Sp}_{T(h)} \rightleftarrows \text{coSp}(\text{CAlg}^{\text{nu}}(\text{Sp}_{T(h)})) : \text{oblv}_{\text{Com}}^{\text{co}}.$$

The right adjoint exists by [8, Proposition 3.1.13(iii)]. Since both categories are stable, $\text{oblv}_{\text{Com}}^{\text{co}}$ is exact.

Proposition 11.2. *There is a natural equivalence $\mathrm{oblv}_{\mathrm{Lie}} T_\infty \mathrm{res}_h^{\mathrm{bar}} \simeq \mathrm{oblv}_{\mathrm{Com}}^{\mathrm{co}}$.*

Proof. The functor $\mathrm{com}_h^{\mathrm{bar}} U_\infty \mathrm{Free}_{\mathrm{Lie}}$ preserves finite colimits and

$$\mathrm{ev}_0 \mathrm{com}_h^{\mathrm{bar}} U_\infty \mathrm{Free}_{\mathrm{Lie}} \simeq \mathrm{com}_0 \mathrm{pr}_0 U_\infty \mathrm{Free}_{\mathrm{Lie}} \simeq \mathrm{com}_0 = \mathrm{Free}_{\mathrm{Com}}^{\mathrm{nu}}.$$

The costabilization universal property therefore identifies $\mathrm{com}_h^{\mathrm{bar}} U_\infty \mathrm{Free}_{\mathrm{Lie}}$ with $\mathrm{Free}_{\mathrm{Com}}^{\mathrm{co}}$. Their right adjoints are respectively $\mathrm{oblv}_{\mathrm{Lie}} T_\infty \mathrm{res}_h^{\mathrm{bar}}$ and $\mathrm{oblv}_{\mathrm{Com}}^{\mathrm{co}}$, which proves the assertion. $\square$

12. LOCALIZED GROUP RINGS OF CONNECTIVE COVERS

From now on assume $h \geq 2$. Work on $\mathrm{thick}_{\mathrm{Sp}(p)}(K(1))$, the smallest full stable subcategory of p -local spectra containing $K(1)$ and closed under retracts. Every object of $\mathrm{thick}_{\mathrm{Sp}(p)}(K(1))$ is rationally acyclic.

Let d_h be the connectivity bound for Bousfield's unstable finite localization L_h^f in [3, §4.3 and Corollary 4.8]. Choose

$$c \geq \max\{2, d_h + 1\}.$$

For a connective p -local spectrum P , define the localized augmentation-ideal group ring by

$$\mathrm{AugGrp}_h(P) = I\left(L_{T(h)} \Sigma_+^\infty \Omega^\infty P\right).$$

The multiplication is induced by the additive grouplike $\mathbb{E}_\infty$ -space $\Omega^\infty P$, and the augmentation is induced by its map to the point. Consider the functor

$$(12.1) \quad \mathrm{thick}_{\mathrm{Sp}(p)}(K(1)) \longrightarrow \mathrm{CAlg}^{\mathrm{nu}}(\mathrm{Sp}_{T(h)}), \quad M \longmapsto \mathrm{AugGrp}_h(\tau_{\geq c} M).$$

Proposition 12.1. *The functor in (12.1) preserves finite colimits. It therefore admits an exact lift*

$$\mathrm{AugGrp}_{h,c}^{\mathrm{co}}: \mathrm{thick}_{\mathrm{Sp}(p)}(K(1)) \longrightarrow \mathrm{coSp}(\mathrm{CAlg}^{\mathrm{nu}}(\mathrm{Sp}_{T(h)})),$$

with terms

$$(\mathrm{AugGrp}_{h,c}^{\mathrm{co}} M)_n \simeq \mathrm{AugGrp}_h(\tau_{\geq c} \Sigma^{-n} M).$$

Connective truncation does not preserve pushouts. Lemma 12.2 shows that sufficiently connected bounded Postnikov errors disappear under AugGrp_h . Lemma 12.3 identifies the error in a truncated pushout as a single Eilenberg–MacLane spectrum. These two assertions, together with colimit preservation by group rings, prove the proposition.

Lemma 12.2. *Let $f: P \rightarrow P'$ be a map of c -connective, rationally acyclic, p -local spectra. If $\mathrm{cofib}(f)$ has homotopy in only finitely many degrees, then $\mathrm{AugGrp}_h(f)$ is an equivalence.*

Proof. For each $i \geq 1$, mapping a finite type- i spectrum into $\mathrm{cofib}(f)$ gives a bounded spectrum. Inverting a positive-degree v_i self-map annihilates it: in each fixed degree, the defining sequential colimit eventually samples degrees outside the bounded range. Thus f is a stable v_i -periodic equivalence for every $i \geq 1$.

Bousfield's comparison [3, Proposition 5.1] implies that $\Omega^\infty f$ is an unstable v_i -periodic equivalence for every such i . The spaces are connected nilpotent infinite loop spaces. Their positive homotopy groups are rationally zero, so they are rationally contractible. Thus $\Omega^\infty f$ is also a height-zero periodic equivalence. Both spaces are d_h -connected by our choice of c . It follows from [3, Corollary 4.8] that $\Omega^\infty f$ is an L_h^f -equivalence.

By [3, Lemma 5.11], the map $X \rightarrow L_h^f X$ is an $S(h)$ -homology equivalence. The Bousfield class of $S(h)$ is the join of those of $T(0), \dots, T(h)$ [3, §6.12]. Naturality and two-out-of-three show that every L_h^f -equivalence is a $T(h)$ -homology equivalence. Therefore $L_{T(h)} \Sigma_+^\infty \Omega^\infty f$ is an equivalence, as is its augmentation ideal. $\square$

Lemma 12.3. *Suppose $M = M_1 \amalg_{M_0} M_2$ is a pushout of spectra, and set*

$$P = \tau_{\geq c} M_1 \amalg_{\tau_{\geq c} M_0} \tau_{\geq c} M_2.$$

The natural map $P \rightarrow \tau_{\geq c} M$ has cofiber $\Sigma^c H(V)$, where

$$V = \ker(\pi_{c-1} M_0 \rightarrow \pi_{c-1} M_1 \oplus \pi_{c-1} M_2).$$

Proof. The subcategory of c -connective spectra is closed under colimits, so P is c -connective and its map to M factors through $\tau_{\geq c} M$. Compare the long exact sequences for

$$M_0 \rightarrow M_1 \oplus M_2 \rightarrow M$$

and the corresponding sequence of connective covers with cofiber P . The comparison is an isomorphism on π_k for $k > c$, and an injection on π_c with cokernel V . Both spectra vanish below c , proving the assertion. $\square$

Proof of proposition 12.1. The spherical group-ring functor

$$\Sigma_+^\infty \Omega^\infty : \mathrm{Sp}_{\geq 0} \rightarrow \mathrm{CAlg}(\mathrm{Sp})$$

is left adjoint to gl_1 [1, Theorem 5.1 and Remark 5.2, published version]; hence it preserves colimits. The canonical augmentation comes from $P \rightarrow 0$. Since colimits in the slice over the sphere are created in commutative algebras, this augmentation lift also preserves colimits. The same conclusion holds after restricting to connective p -local spectra.

The class of $T(h)$ -equivalences is closed under smash product. Thus localization is symmetric monoidal and induces a colimit-preserving functor on commutative algebras. Taking augmentation ideals is an equivalence with nonunital algebras, so it preserves colimits as well. Consequently AugGrp_h preserves colimits on connective spectra.

For a pushout $M = M_1 \amalg_{M_0} M_2$ in $\mathrm{thick}_{\mathrm{Sp}_{(p)}}(K(1))$, use the spectrum P of lemma 12.3. It and $\tau_{\geq c} M$ are rationally acyclic and c -connective, and their comparison has bounded cofiber. Lemma 12.2 gives

$$\begin{aligned} \mathrm{AugGrp}_h(\tau_{\geq c} M_1) \amalg_{\mathrm{AugGrp}_h(\tau_{\geq c} M_0)} \mathrm{AugGrp}_h(\tau_{\geq c} M_2) &\simeq \mathrm{AugGrp}_h(P) \\ &\simeq \mathrm{AugGrp}_h(\tau_{\geq c} M). \end{aligned}$$

Also $\mathrm{AugGrp}_h(\tau_{\geq c} 0) = 0$. Therefore $\mathrm{AugGrp}_h \circ \tau_{\geq c}$ preserves finite colimits. Its lift follows from the costabilization universal property. Explicitly, finite-colimit preservation gives

$$\Sigma_{\mathrm{CAlg}^{\mathrm{nu}}(\mathrm{Sp}_{T(h)})} \mathrm{AugGrp}_h(\tau_{\geq c} \Sigma^{-n-1} M) \simeq \mathrm{AugGrp}_h(\tau_{\geq c} \Sigma^{-n} M),$$

which supplies the cospectrum structure. Since both source and target of the lift are stable, the lift is exact. $\square$

13. A NONZERO TOWER IN THE KERNEL

We prove the second part of theorem 1.2 by taking

$$(13.1) \quad A = \mathrm{res}_h^{\mathrm{bar}} \mathrm{AugGrp}_{h,c}^{\mathrm{co}}(K(1)) \in \mathrm{BarTower}_h.$$

Propositions 11.1 and 12.1 give its terms as in (1.2) and its compatible structure equivalences. It remains to establish $\mathrm{oblv}_{\mathrm{Lie}} T_\infty(A) = 0$ and $A_0 \neq 0$.

Lemma 13.1. *An exact functor between stable p -local categories commutes with the action of finite p -local spectra. A finite p -local spectrum V of type h acts conservatively on $\mathrm{Sp}_{T(h)}$ by tensor product.*

Proof. Exact functors commute with finite colimits, suspensions, desuspensions, and retracts. The compatibility with finite spectra follows by their universal generation from the sphere under these operations. Inverting maps of degree prime to p gives the same compatibility for finite p -local spectra.

For the second assertion choose a telescope $T(h)$ of a v_h self-map of a finite type- h spectrum. Smashing its defining sequence with V gives the telescope of a v_h self-map on another finite type- h spectrum. Telescope Bousfield classes are independent of the choice of such a spectrum and self-map [3, §3]. Hence

$$\langle T(h) \wedge V \rangle = \langle T(h) \rangle.$$

If $L_{T(h)}(V \wedge Z) = 0$, it follows that $T(h) \wedge Z = 0$. When Z is $T(h)$ -local this implies $Z = 0$. Applying the zero-object assertion to cofibers proves conservativity on morphisms. $\square$

Proposition 13.2. *The object in (13.1) satisfies $\text{oblv}_{\text{Lie}} T_\infty(A) = 0$.*

Proof. Choose a finite p -local spectrum V of type h . Since $h \geq 2$, its $K(1)$ -homology is zero, and therefore $V \wedge K(1) = 0$. The category $\text{thick}_{\text{Sp}(p)}(K(1))$ is closed under finite p -local spectral actions. The composite

$$\text{oblv}_{\text{Com}}^{\text{co}} \text{AugGrp}_{h,c}^{\text{co}} : \text{thick}_{\text{Sp}(p)}(K(1)) \longrightarrow \text{Sp}_{T(h)}$$

is exact. Thus lemma 13.1 gives

$$V \otimes \text{oblv}_{\text{Com}}^{\text{co}} \text{AugGrp}_{h,c}^{\text{co}}(K(1)) \simeq \text{oblv}_{\text{Com}}^{\text{co}} \text{AugGrp}_{h,c}^{\text{co}}(V \wedge K(1)) = 0.$$

Conservativity of $V \otimes -$ on $\text{Sp}_{T(h)}$ implies $\text{oblv}_{\text{Com}}^{\text{co}} \text{AugGrp}_{h,c}^{\text{co}}(K(1)) = 0$. Proposition 11.2 now identifies this object with $\text{oblv}_{\text{Lie}} T_\infty(A)$. $\square$

Proposition 13.3. *For every $h \geq 2$, the zeroth term of the tower in (13.1) is nonzero.*

Proof. Put

$$X = \Omega^\infty K(1), \quad Y = \Omega^\infty \tau_{\geq c} K(1).$$

Bousfield observes immediately after [3, Theorem 6.5] that X is not virtually $K(h)$ -acyclic for any $h \geq 2$. We use the virtual-equivalence properties in [3, Theorem 6.3] to pass to the specified connected cover.

If Y were $K(h)$ -acyclic, then $Y \rightarrow *$ would be a $K(h)$ -homology equivalence of connected H -spaces and hence a virtual $K(h)$ -equivalence, by part (i) of that theorem. The connected-cover map $Y \rightarrow X$ is a virtual $K(h)$ -equivalence by part (ii). Its two-out-of-three property, part (iii), would make X virtually $K(h)$ -acyclic, a contradiction. Thus

$$\widetilde{K}(h)_*(Y) \neq 0.$$

The basepoint of Y splits the augmentation of its suspension spectrum, giving

$$\text{oblv AugGrp}_h(\tau_{\geq c} K(1)) \simeq L_{T(h)} \Sigma^\infty Y.$$

If this vanished, the collapse $\Sigma_+^\infty Y \rightarrow S_{(p)}$ would be a stable v_h -periodic equivalence. Such a map is a $K(h)$ -homology equivalence by [3, Theorem 6.2], contrary to the preceding nonvanishing. Therefore $\text{oblv AugGrp}_h(\tau_{\geq c} K(1)) \neq 0$. Since

$$A_0 = \text{res}_{\mathbb{E}_0}^{\mathbb{E}_\infty} \text{AugGrp}_h(\tau_{\geq c} K(1)) \simeq \text{oblv AugGrp}_h(\tau_{\geq c} K(1)),$$

the assertion follows. $\square$

Proof of theorem 1.2(ii). For the tower A in (13.1), Propositions 13.2 and 13.3 give $\text{oblv}_{\text{Lie}} T_\infty(A) = 0$ and $A \neq 0$. If $A \simeq U_\infty L$, full faithfulness of U_∞ would imply

$$0 \simeq \text{oblv}_{\text{Lie}} T_\infty(A) \simeq \text{oblv}_{\text{Lie}} T_\infty U_\infty L \simeq \text{oblv}_{\text{Lie}} L.$$

The forgetful functor on Lie algebras is conservative, so $L = 0$ and hence $A = 0$, a contradiction. Thus $A \notin \text{EssIm}(U_\infty)$. $\square$

APPENDIX A. A DIRECT PROOF AT HEIGHT ZERO

We give a direct proof of the rational equivalence by computing suspension in the tower category and the adjunction unit on trivial Lie algebras.

Set $T(0) = H\mathbb{Q}$ and work in $\text{Sp}_{\mathbb{Q}} \simeq D(\mathbb{Q})$. Define

$$\text{BarTower}_0 = \varprojlim_{n \geq 0} (\text{Alg}_{\mathbb{E}_n}^{\text{nu}}(D(\mathbb{Q})), B_{n+1})$$

using the actual reduced bar constructions. The spectral normalization of Lie is retained: its rational binary bracket has homological degree -1 .

Theorem A.1. *The actual higher enveloping functor*

$$U_\infty: \text{Alg}_{\text{Lie}}(\text{Sp}_{\mathbb{Q}}) \xrightarrow{\simeq} \text{BarTower}_0$$

is an equivalence.

We first show that suspension of any rational bar tower is a free tower. Together with conservativity of loops, this makes $\text{oblv}_{\text{Lie}} T_\infty$ conservative. We then prove that the unit of $U_\infty \dashv T_\infty$ is an equivalence, by reducing it to trivial Lie algebras and computing their actual bar transitions.

For a reduced operad $\mathcal{O}$, write $\text{indec}_{\mathcal{O}}$ for derived indecomposables and $\text{triv}_{\mathcal{O}}$ for the trivial-algebra functor. As in the main text, let $F(L) = \mathbb{Q} \oplus \text{indec}_{\text{Lie}}(L)$ be the underlying augmented Chevalley–Eilenberg complex. Over $\mathbb{Q}$, its usual complex gives a coherent symmetric monoidal equivalence

$$F(L \times M) \simeq F(L) \otimes F(M).$$

Indeed, the symmetric-algebra decomposition for a direct sum respects the Chevalley–Eilenberg differential, since the mixed brackets vanish. The Chevalley–Eilenberg complex is formed by direct sums of symmetric powers, so the decomposition applies to unbounded complexes.

Consequently, the actual construction of [8, §2.2.13] applies over $\mathbb{Q}$. Namely, apply F to the canonical $\mathbb{E}_n$ -algebra object given by the n -fold loop object of L in Lie algebras, and then take the augmentation ideal. These functors U_n commute with actual bar and give an adjunction $U_\infty \dashv T_\infty$. The free-algebra calculation is

$$(A.1) \quad \text{Free}_{\text{bar}} := U_\infty \text{Free}_{\text{Lie}}, \quad (\text{Free}_{\text{bar}} E)_n \simeq \text{Free}_{\mathbb{E}_n}^{\text{nu}}(\Sigma^{-n} E).$$

The right adjoint of U_n has underlying complex $\Sigma^n \text{oblv}_{\mathbb{E}_n}$: its construction is n -fold bar in Lie algebras, whose underlying additive bar is n -fold suspension. Taking left adjoints gives (A.1) and its comparison maps, as in the proof of [8, Proposition 2.2.14]. In particular,

$$(A.2) \quad \text{oblv}_{\text{Lie}} T_\infty(A) \simeq \varprojlim_n \Sigma^n \text{oblv}_{\mathbb{E}_n}(A_n).$$

The transition maps in this limit are the mates of the bar-free comparisons in (A.1).

Lemma A.2. *For a reduced differential graded operad $\mathcal{O}$ over $\mathbb{Q}$ with $\mathcal{O}(1) = \mathbb{Q}$, there are natural equivalences*

$$\Omega_{\text{Alg}_{\mathcal{O}}} A \simeq \text{triv}_{\mathcal{O}}(\Sigma^{-1} \text{oblv}_{\mathcal{O}} A), \quad \Sigma_{\text{Alg}_{\mathcal{O}}} \simeq \text{Free}_{\mathcal{O}} \Sigma \text{indec}_{\mathcal{O}}.$$

They are compatible with operad restriction and operadic degree shifts.

Proof. Use cohomological complexes for this calculation. The polynomial de Rham algebra $\mathbb{Q}[t, dt]$, with $|t| = 0$, $|dt| = 1$, and $d(t) = dt$, gives a path object $A \otimes \mathbb{Q}[t, dt]$. The endpoint map is degreewise surjective and the inclusion of constant paths is a quasi-isomorphism. Hence the loop object is represented by $A \otimes J$, where

$$J = \{f(t) + g(t)dt : f(0) = f(1) = 0\}.$$

The nonunital commutative differential graded algebra map

$$\mathbb{Q}e \longrightarrow J, \quad |e| = 1, \quad e^2 = 0, \quad e \longmapsto dt$$

is a quasi-isomorphism: $H^0(J) = 0$, and integration identifies $H^1(J)$ with $\mathbb{Q}$. Tensoring with an arbitrary complex preserves this quasi-isomorphism. All operations of arity at least two on $A \otimes \mathbb{Q}e$ vanish. This proves the loop formula, and taking left adjoints proves the suspension formula.

The same path algebra, endpoint maps, and inclusion $e \mapsto dt$ work for every $\mathcal{O}$. They commute with operad restriction, and with operadic shifts under the suspension sign isomorphisms. These are compatibilities of natural transformations, so also hold for composites and pass to their mates. $\square$

Proposition A.3. *For every $A \in \text{BarTower}_0$, there is a natural equivalence of compatible towers*

$$(A.3) \quad \Sigma_{\text{BarTower}_0} A \simeq \text{Free}_{\text{bar}}(\Sigma A_0).$$

The functor $\text{oblv}_{\text{Lie}} T_\infty$ is conservative.

Proof. Let ω_{n+1} be the right adjoint of actual bar B_{n+1} . By [5, Theorem 3.11], it is restriction along the operad map constructed there from actual bar, followed by the underlying shift Σ^{-1} . In particular,

$$\omega_{n+1} \text{triv}_{\mathbb{E}_n} \simeq \text{triv}_{\mathbb{E}_{n+1}} \Sigma^{-1}, \quad \text{oblv}_{\mathbb{E}_{n+1}} \omega_{n+1} \simeq \Sigma^{-1} \text{oblv}_{\mathbb{E}_n}.$$

Taking left adjoints gives coherent comparisons

$$\text{indec}_{\mathbb{E}_n} B_{n+1} \simeq \Sigma \text{indec}_{\mathbb{E}_{n+1}}, \quad B_{n+1} \text{Free}_{\mathbb{E}_{n+1}}^{\text{nu}} \simeq \text{Free}_{\mathbb{E}_n}^{\text{nu}} \Sigma.$$

Since $\text{indec}_{\mathbb{E}_0} = \text{id}$, a compatible tower therefore satisfies $\text{indec}_{\mathbb{E}_n}(A_n) \simeq \Sigma^{-n} A_0$.

The loop comparison of lemma A.2 commutes with ω_{n+1} : on the path models, both composites are the same map induced by $e \mapsto dt$ on the shifted underlying complex. Taking mates identifies the canonical comparison $B_{n+1} \Sigma \simeq \Sigma B_{n+1}$ with the comparison obtained from the two displayed free and indecomposable identities. Thus the suspension formulas of lemma A.2 respect the actual tower maps and their composites. Suspension in the tower category is computed levelwise, so

$$(\Sigma_{\text{BarTower}_0} A)_n \simeq \text{Free}_{\mathbb{E}_n}^{\text{nu}}(\Sigma \text{indec}_{\mathbb{E}_n} A_n) \simeq \text{Free}_{\mathbb{E}_n}^{\text{nu}}(\Sigma^{1-n} A_0),$$

with precisely the structure maps of (A.3).

The proof of proposition 9.1 applies to $D(\mathbb{Q})$ as well: it uses only the actual bar–suspension adjunctions and restriction of operadic structures. Hence loops on BarTower_0 are conservative. If a morphism f becomes an equivalence under $\text{oblv}_{\text{Lie}} T_\infty$, then for every tower X , adjunction and (A.3) identify

$$\text{Map}(X, \Omega f) \simeq \text{Map}(\Sigma X, f) \simeq \text{Map}(\text{Free}_{\text{bar}}(\Sigma X_0), f)$$

as an equivalence of mapping spaces. Yoneda makes Ωf an equivalence, and conservativity of loops makes f an equivalence. $\square$

Lemma A.4. *For a rational complex V , the natural suspension-to-bar map, followed by the free commutative bar equivalence, is*

$$\Sigma \text{Free}_{\text{Com}}^{\text{nu}}(V) \longrightarrow B \text{Free}_{\text{Com}}^{\text{nu}}(V) \simeq \text{Free}_{\text{Com}}^{\text{nu}}(\Sigma V).$$

It is the identity on weight one and zero on every higher weight, under the weight decompositions of source and target.

Proof. Write $q : \text{Free}_{\text{Com}}^{\text{nu}}(V) \rightarrow V$ for projection to weight one. The reduced shuffle bar complex has a natural map to $\text{Free}_{\text{Com}}^{\text{nu}}(\Sigma V)$ given on a word of length k by

$$[a_1 | \cdots | a_k] \mapsto \frac{1}{k!} (\Sigma q(a_1)) \cdots (\Sigma q(a_k)),$$

with the suspension signs in the bar convention. It is a chain map because $q(ab) = 0$, and it respects shuffle multiplication. It is a quasi-isomorphism by the free commutative Koszul resolution: for an even generator the resolution has an exterior suspended generator, and for an odd generator it has a polynomial suspended generator with differential sending it to the original generator. The latter complex is exact in positive resolution degrees because every positive integer is invertible in $\mathbb{Q}$. Tensoring these resolutions proves the assertion for a finite graded set of generators, and filtered colimits extend it to arbitrary graded vector spaces. Every complex over a field is quasi-isomorphic to its homology, so invariance under quasi-isomorphisms gives the assertion for every complex V .

Both the symmetric-algebra functor and bar preserve quasi-isomorphisms, since tensor products over $\mathbb{Q}$ and geometric realizations preserve equivalences. Moreover, the canonical free commutative bar map in the opposite direction is generated by the words $[v]$. Its composite with the displayed map is the identity on generators, and hence on the free commutative algebra. The quasi-isomorphism is therefore the canonical free commutative bar equivalence.

The suspension-to-bar map is inclusion of the words of length one. Its composite with this equivalence is Σq followed by inclusion of the generators, proving the formula on the direct sum of weights. $\square$

Proposition A.5. *The rational actual higher enveloping functor U_∞ is fully faithful.*

Proof. Let $\eta_L : L \rightarrow T_\infty U_\infty L$ be the adjunction unit. We first compute it when $L = \text{triv}_{\text{Lie}} E$. The trivial Lie functor preserves limits, so the canonical n -fold loop object of L is obtained by applying it to the additive n -fold loop object of E . The Chevalley–Eilenberg functor sends this additive structure to multiplication on the symmetric algebra. Thus, as a compatible actual bar tower,

$$(A.4) \quad U_n(\text{triv}_{\text{Lie}} E) \simeq \text{res}_{\mathbb{E}_n}^{\mathbb{E}_\infty} \text{Free}_{\text{Com}}^{\text{nu}}(\Sigma^{-n} E).$$

This includes the bar comparisons: the symmetric-algebra functor preserves colimits, so it carries the additive bar of $\Sigma^{-n-1} E$ to the commutative bar of its free algebra.

Apply (A.2) to the tower (A.4). Its transitions are the suspended suspension-to-bar maps of lemma A.4. Their weight-one summands are the constant tower E , and their remaining summands have zero transition maps. The inverse limit of the latter tower is zero: in the product-fiber formula for a sequential limit, $1 - \text{shift}$ is the identity on this part. Consequently

$$\text{oblv}_{\text{Lie}} T_\infty U_\infty(\text{triv}_{\text{Lie}} E) \simeq E.$$

At stage zero the unit is the unit of $\text{indec}_{\text{Lie}} \dashv \text{triv}_{\text{Lie}}$. Its counit on E is the weight-one projection $\text{Free}_{\text{Com}}^{\text{nu}}(E) \rightarrow E$. The triangle identity says that this projection composed with the underlying unit is id_E . Since projection to weight one identifies the inverse limit with E , the unit $\eta_{\text{triv}_{\text{Lie}} E}$ is an equivalence.

For general L , the actual construction gives

$$(\text{oblv}_{\text{Lie}} T_\infty U_\infty)(L) \simeq \varprojlim_n \Sigma^n \text{indec}_{\text{Lie}}(\Omega_{\text{Alg}_{\text{Lie}}}^n L).$$

The transitions are the suspension mates of the canonical maps

$$\text{indec}_{\text{Lie}}(\Omega_{\text{Alg}_{\text{Lie}}} L) \longrightarrow \Sigma^{-1} \text{indec}_{\text{Lie}}(L).$$

Indeed, the mate of the actual bar comparison is the inclusion of the one-simplex part of bar, followed by the internal bar-loop counit. After applying the colimit-preserving functor $\text{indec}_{\text{Lie}}$, this is precisely the mate of the displayed map. The same construction identifies the unit at each stage with the suspended unit on $\Omega_{\text{Alg}_{\text{Lie}}}^n L$.

Reindexing the inverse tower consequently gives the canonical, unit-compatible equivalence

$$(\mathrm{oblv}_{\mathrm{Lie}} T_\infty U_\infty)(\Omega_{\mathrm{Alg}_{\mathrm{Lie}}} L) \simeq \Sigma^{-1}(\mathrm{oblv}_{\mathrm{Lie}} T_\infty U_\infty)(L).$$

This recovers the loop calculation in [8, proof of Proposition 2.3.11] by symmetric monoidality and reindexing. By lemma A.2, $\Omega_{\mathrm{Alg}_{\mathrm{Lie}}} L$ is trivial. Its unit is therefore an equivalence, so the unit-compatible comparison shows that $\Sigma^{-1} \mathrm{oblv}_{\mathrm{Lie}}(\eta_L)$ is an equivalence. Spectral desuspension and the forgetful functor are conservative, proving the assertion for every L . $\square$

Proof of theorem A.1. Full faithfulness follows from proposition A.5. The triangle identity then shows that T_∞ sends the counit $U_\infty T_\infty A \rightarrow A$ to an equivalence. By proposition A.3, $\mathrm{oblv}_{\mathrm{Lie}} T_\infty$ is conservative. The counit is therefore an equivalence for every A , proving essential surjectivity. $\square$

Remark A.6. Shi’s thesis [9, Theorem 6.2.1.17, pp. 181–182] proves the rational equivalence for the transition functors defined by Koszul duality in its Construction 6.1.0.2. Its Corollary 6.1.0.14 identifies those transitions with actual bar. In [8], Theorem 2.4.12 proves the Koszul-transition equivalence, while Conjecture 2.4.10 asks for the coherent comparison of the two constructions. Our theorem 7.2 proves this comparison, giving the short proof of the rational actual-bar equivalence in section 8. The direct argument above also gives the free-suspension formula (A.3). A related comparison is announced in [2, Remark 3.2], with its proof deferred there to further work.

Proposition A.7. *For every prime p , every $c \geq 2$, and every $n \geq 0$,*

$$I(L_{\mathbb{Q}} \Sigma_+^\infty \Omega^\infty \tau_{\geq c} \Sigma^{-n} K(1)) \simeq 0.$$

Thus the terms of the higher-height counterexample all vanish when its localization is replaced by rational localization.

Proof. The space $Y = \Omega^\infty \tau_{\geq c} \Sigma^{-n} K(1)$ is simply connected, and its homotopy groups are torsion. The rational Whitehead theorem therefore gives $\tilde{H}_*(Y; \mathbb{Q}) = 0$, or equivalently $L_{\mathbb{Q}} \Sigma^\infty Y = 0$. This is the underlying spectrum of the rationalized augmentation ideal in the statement. Conservativity of the forgetful functor proves the claim. $\square$

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